



OPEN Deep learning-based battery health prediction for enhancing electric vehicle performance

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Reliable and sustainable battery diagnostics are essential for advancing electric vehicle (EV) technologies and fulfilling Sustainable Development Goal 7 (SDG 7): Affordable and Clean Energy. This study proposes a hybrid deep learning framework that integrates one-dimensional Convolutional Neural Networks (1D-CNNs), Temporal Convolutional Networks (TCNs), and Long Short-Term Memory (LSTM) layers, along with an attention mechanism, for intelligent EV battery health diagnostics. Differential Voltage (dV/dQ), Differential Current (dI/dV), and Incremental Capacity Analysis (ICA, dQ/dV) features were extracted and denoised from over 10,000 charge–discharge cycles sourced from the NASA PCoE, Oxford, and CALCE battery degradation datasets. The proposed model achieved a State-of-Health (SOH) prediction accuracy of $R^2 = 0.983$ and RMSE = 0.021, outperforming conventional CNN, LSTM, and XGBoost baselines by up to 25% in accuracy. With only 0.35 million parameters, the model demonstrated an average inference latency of 6.1 ms and energy consumption of 0.63 mJ per sample—a 27% reduction compared to Transformer-based architectures. These results confirm the framework's robustness, scalability, and real-time feasibility for embedded Battery Management Systems (BMS). By improving diagnostic precision, extending battery lifespan, and reducing computational energy demand, the proposed method directly contributes to sustainable mobility and the broader goals of energy efficiency and the circular economy under SDG 7.

Keywords Electric vehicles, Battery diagnostics, Incremental capacity analysis, Differential voltage, Deep learning, SDG 7, Sustainability

The global transition toward sustainable transportation systems depends heavily on the performance, reliability, and lifespan of lithium-ion batteries in electric vehicles (EVs). Battery health degradation leads to range anxiety, reduced energy efficiency, and increased life-cycle costs, posing a significant barrier to achieving Sustainable Development Goal 7 (SDG 7): Affordable and Clean Energy. Hence, intelligent, data-driven diagnostic tools are essential for enhancing energy storage reliability, promoting circular-economy practices, and supporting the United Nations Sustainable Development Agenda^{1,2}. Conventional methods for battery health monitoring—such as equivalent circuit models (ECMs), empirical curve fitting, or Incremental Capacity Analysis (ICA)—offer limited adaptability to nonlinear degradation behaviors under varying operating conditions^{3,4}. Recently, deep learning has emerged as a powerful tool for modeling complex temporal–electrochemical relationships from raw sensor data, enabling accurate State-of-Health (SOH) estimation and Remaining Useful Life (RUL) prediction. This paper introduces a hybrid deep learning framework that combines CNN, TCN, and LSTM layers with attention mechanisms to analyze Differential Voltage (DV), Differential Current (DC), and ICA features for sustainable EV battery diagnostics^{5–7}.

The accelerating global shift toward electrified transportation has become a cornerstone of the energy transition and the broader United Nations Sustainable Development Goals (SDGs). Among these, Sustainable

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Development Goal 7 (SDG 7) - *Affordable and Clean Energy* - emphasizes access to reliable, sustainable, and modern energy services. Electric vehicles (EVs) play a crucial role in achieving this target by reducing dependence on fossil fuels, mitigating greenhouse gas emissions, and improving urban air quality. However, the widespread adoption of EVs critically depends on the performance, safety, and longevity of lithium-ion batteries (LIBs), which currently dominate the electrochemical energy storage landscape^{8,9}. Battery health degradation is an unavoidable consequence of repetitive charge–discharge cycling, temperature fluctuations, and side reactions such as solid-electrolyte interphase (SEI) formation and lithium plating^{10,11}. These degradation processes lead to capacity fade, power loss, and increased internal resistance, ultimately reducing the range and reliability of EVs. Accurate estimation of the *State of Health* (SOH) and prediction of the *Remaining Useful Life* (RUL) are therefore vital for maintaining battery safety, enabling predictive maintenance, and ensuring efficient energy use across the EV life cycle. These capabilities directly support SDG 7 by extending the lifetime of energy storage systems and promoting sustainable mobility^{12,13}.

Battery diagnostics and prognostics have evolved rapidly over the past decade. Conventional model-based techniques such as Equivalent Circuit Models (ECM) and Electrochemical Impedance Spectroscopy (EIS) remain valuable for identifying degradation mechanisms and estimating internal parameters^{14,15}. However, they require extensive calibration, are sensitive to temperature and aging, and are often impractical for large-scale or real-time applications¹⁶. Empirical techniques such as Incremental Capacity Analysis (ICA) and Differential Voltage Analysis (DVA) are widely used to infer degradation via peak shifts and area variations in the dQ/dV and dV/dQ curves^{17,18}. Yet, these methods remain susceptible to measurement noise and are limited by their reliance on complete charge–discharge cycles, which may not always be feasible in real-world EV operation⁸. In recent years, machine learning (ML) and deep learning (DL) have transformed battery diagnostics by learning complex nonlinear mappings between electrochemical signals and health indicators^{8,19,20}. Shallow ML models such as Support Vector Machines (SVM), Random Forests (RF), and Gradient Boosting (XGBoost) have achieved reasonable SOH prediction accuracy when combined with handcrafted ICA or voltage-based features²¹. However, these models require extensive feature engineering and often lack generalization capability across different battery chemistries and operating regimes.

Deep neural networks, in contrast, automatically extract multiscale representations from raw sensor data. Convolutional Neural Networks (CNNs) excel at identifying local electrochemical features such as ICA peak distortions, while Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), capture long-term temporal correlations across cycles^{22,23}. Temporal Convolutional Networks (TCNs) and Transformers have recently shown promise for modeling long-range dependencies with reduced computational overhead²⁴. Furthermore, physics-informed neural networks (PINNs) and hybrid approaches that fuse ECM parameters with learned representations have emerged as powerful tools to enforce physical consistency and interpretability^{14,25}. Despite these advances, challenges persist in ensuring model explainability, deployment efficiency, and sustainability — particularly for embedded battery management systems (BMS) where power, latency, and interpretability are critical^{26,27}. Although Remaining Useful Life (RUL) prediction is an essential objective in battery management, this study focuses on State-of-Health (SOH) estimation as the primary, experimentally validated task. References to RUL are included only to indicate a potential extension of the proposed framework, as a rigorous RUL study would require explicitly defined prediction horizons, dedicated evaluation metrics, and benchmark protocols that are beyond the scope of the present work.

Recent advances in electric vehicle (EV) charging infrastructure have underscored the need for robust integration with renewable energy sources, intelligent grid interaction, and scalable optimization techniques to support rapid EV adoption and carbon-reduction goals. Systematic reviews highlight the increasing focus on coordinated charging strategies, machine learning applications, and strategic station placement to improve infrastructure resilience and sustainability²⁸. Grid-integrated solutions that combine renewable generation and battery storage systems reduce operational costs and emissions while enhancing grid stability²⁹. Studies also note the critical role of AI-driven optimization frameworks in managing dynamic demand and grid constraints, particularly under variable renewable generation³⁰. Furthermore, comprehensive analyses of EV charging infrastructure underscore both the opportunities and challenges associated with renewable energy integration, power-system impacts, and smart-charging protocols^{31,32}.

Despite substantial advancements in battery health diagnostics, key limitations persist in existing studies. Many existing models fail to integrate multi-domain electrochemical features—such as Differential Voltage (dV/dQ), Differential Current (dI/dV), and Incremental Capacity (dQ/dV)—that collectively capture the full degradation spectrum^{17,18}. Moreover, traditional RNN-based and CNN–LSTM frameworks often exhibit weak long-horizon temporal retention, limiting their robustness under extended cycling and dynamic load conditions^{12,33}. Although deep learning approaches can achieve high predictive accuracy, they frequently sacrifice interpretability, providing limited insight into the underlying electrochemical processes and reducing trustworthiness in safety-critical EV applications^{9,13}. In addition, the growing computational complexity of deep architectures increases energy consumption, conflicting with sustainability objectives aligned with Sustainable Development Goal 7 (SDG 7).

Motivated by these limitations, this work introduces a deployment-oriented hybrid diagnostic framework that unifies multi-domain electrochemical features within a principled preprocessing and feature-alignment pipeline, and integrates a lightweight CNN–TCN–LSTM architecture enhanced by attention mechanisms. While the individual learning components are well established, their system-level integration for real-time, energy-efficient Battery Management Systems (BMS) remains underexplored. The novelty of the proposed approach lies not in new learning primitives, but in its robust fusion of complementary electrochemical representations, explicit efficiency-aware design, and cross-dataset validation, enabling accurate, interpretable, and low-energy State-of-Health (SOH) estimation suitable for embedded EV BMS deployment. By prioritizing deployability

and sustainability alongside predictive performance, this study bridges a critical gap between laboratory-scale battery health models and practical, real-world diagnostic systems.

Research Objectives: The present study aims to design a sustainable, interpretable, and high-accuracy diagnostic framework for EV lithium-ion batteries by leveraging recent advances in deep learning and signal analysis. The primary objectives are:

1. To extract and preprocess multi-domain features, including differential voltage (dV/dQ), differential current (dI/dV), and incremental capacity (dQ/dV) profiles from raw cycling data.
2. To develop a hybrid deep learning model integrating 1D Convolutional Neural Networks (CNN), Temporal Convolutional Networks (TCN), and Long Short-Term Memory (LSTM) networks enhanced with attention mechanisms for joint SOH estimation and RUL prediction.
3. To evaluate the proposed architecture on publicly available datasets (NASA PCoE, Oxford Battery Degradation, and CALCE) and benchmark against state-of-the-art models such as XGBoost, CNN-LSTM, Transformer, and PINN.
4. To ensure the model's interpretability, energy efficiency, and deployability within embedded BMS platforms in line with SDG 7 sustainability principles.

The proposed framework represents a deployment-oriented, system-level integration of established deep learning components (CNN, TCN, LSTM, and attention) tailored for EV battery health diagnostics, rather than a fundamentally new learning paradigm. Accordingly, statements suggesting algorithmic or theoretical novelty were removed or rephrased, and the emphasis was shifted toward the work's practical engineering contributions, including robust multi-domain feature learning, cross-dataset generalization, interpretability validation, and computational efficiency for embedded Battery Management System deployment.

Research Contributions: To distinguish deployment-oriented engineering contributions (robustness, efficiency, and generalization) from claims of algorithmic novelty, the contributions of this work are summarized as follows:

- **Deployment-Oriented Hybrid Architecture:** We engineer a lightweight CNN–TCN–LSTM model with attention that captures both local electrochemical signatures and long-horizon degradation dynamics from multi-domain inputs (DV, DC, and ICA), with design targets aligned to real-time Battery Management System (BMS) constraints.
- **Principled Multi-Domain Feature Synergy:** We present a unified preprocessing, denoising, and feature-alignment pipeline that enables consistent DV/DC/ICA extraction and fusion across heterogeneous datasets, improving robustness to differentiation noise and enhancing cross-cell/cross-chemistry transferability.
- **Physical-Consistency Interpretability:** We use attention visualization and Grad-CAM as *validation tools* to verify that the model emphasizes electrochemically meaningful voltage regions (e.g., degradation-sensitive plateaus), supporting transparent diagnostics without introducing a new explainability method.
- **Energy-Efficient Edge Feasibility:** We explicitly report compactness and on-device efficiency indicators (parameter budget, latency, and inference-energy) and adopt a quantization-ready design to reduce computational overhead and support embedded deployment.
- **Multi-Dataset Benchmarking and Transfer Evaluation:** We evaluate the framework on public datasets (NASA PCoE, Oxford, and CALCE) and benchmark against representative baselines, demonstrating strong performance and transferability (e.g., RMSE = 0.021, $R^2 = 0.983$ in-domain) under protocol and chemistry variation.
- **Sustainability Relevance:** By improving diagnostic precision with low computational demand, the approach supports longer battery service life and reduced premature replacement, contributing to energy-efficiency objectives aligned with SDG 7.

The remainder of this article is organized as follows. Section 2 reviews the latest research in battery diagnostics using both data-driven and hybrid approaches. Section 3 details the proposed hybrid architecture, feature fusion pipeline, and training methodology. Section 4 presents the experimental validation, comparative analysis, and interpretability results. Section 6 concludes with insights into sustainability impacts and future research directions, including the integration of physics-informed neural layers and transfer learning for cross-chemistry generalization. Through this work, we aim to bridge the gap between electrochemical understanding and advanced artificial intelligence, enabling explainable, efficient, and sustainable EV battery diagnostics consistent with the goals of the global Sustainable Development Agenda.

Related work

Recent studies have leveraged artificial intelligence for EV battery prognostics. CNNs have demonstrated success in capturing localized degradation features in charge–discharge curves, whereas recurrent networks, such as LSTMs, effectively model the temporal evolution of SOH³⁴. However, both approaches face limitations in handling long sequences and dynamic features across different chemistries. Hybrid architectures, such as CNN–LSTM or Transformer-based models, have improved generalization but remain computationally expensive for real-time on-board applications³³. Temporal Convolutional Networks (TCNs) provide a scalable solution that offers long receptive fields and low inference latency^{24,35}. Traditional battery diagnostics primarily relied on physics-based and equivalent-circuit modeling frameworks that describe the electrochemical behavior of lithium-ion cells using lumped-parameter systems. Equivalent Circuit Models (ECMs) represent the internal battery dynamics using resistors and capacitors that mimic charge-transfer resistance and double-layer capacitance^{14,36}. Such models enable estimation of the open-circuit voltage (OCV) and internal resistance,

which are key indicators of health. However, ECMs require iterative parameter identification and are sensitive to operating temperature, C-rate, and state of charge (SOC)³⁷. This section describes to explicitly cover recent state-of-the-art battery diagnostic approaches, including DFN-based Physics-Informed Neural Networks (PINNs), federated learning-driven distributed diagnosis frameworks, and lightweight Transformer variants based on knowledge distillation. The revised discussion clarifies how these methods differ from the proposed framework with respect to required physics parameterization, data availability, computational complexity, and deployment feasibility, thereby positioning our work more accurately with respect to contemporary research directions and practical BMS constraints.

Electrochemical impedance spectroscopy (EIS) provides an alternative means to characterize battery health through frequency-domain impedance measurements. Although EIS provides rich diagnostic information on aging mechanisms, such as SEI growth and electrolyte degradation, it requires laboratory-grade instrumentation and stable operating conditions, which limit its use in onboard diagnostics^{10,38}. Empirical data-driven methods, such as Incremental Capacity Analysis (ICA) and Differential Voltage Analysis (DVA), have been widely adopted due to their simplicity and interpretability^{18,39}. ICA examines changes in dQ/dV curves during cycling to identify active material loss and electrode degradation, while DVA assesses dV/dQ variations to analyze overpotential and impedance rise. These methods provide clear physical insights but are prone to noise amplification during differentiation, making them unreliable for partial-cycle data or field applications¹⁸. Several studies have sought to enhance these methods using smoothing filters and curve-fitting techniques. For example, Wang et al.⁴⁰ proposed a denoised ICA technique combined with polynomial fitting, achieving improved SOH correlation under laboratory conditions. Nonetheless, such methods remain insufficiently robust for large-scale EV fleet monitoring, where data are incomplete and operating conditions vary dynamically.

The limitations of empirical and model-based methods have catalyzed significant interest in data-driven diagnostics. Machine learning (ML) models, including Support Vector Machines (SVM), Random Forests (RF), Gaussian Process Regression (GPR), and Gradient Boosting Decision Trees (GBDT), have been applied for both SOH estimation and RUL prediction^{12,21}. These models typically rely on manually engineered features derived from ICA or OCV data. Wang et al.⁴¹ demonstrated that features extracted from early charge curves could predict full cycle life with remarkable accuracy, paving the way for predictive analytics in battery health management. Despite their effectiveness, conventional ML models exhibit limited generalization capability when transferred across chemistries or operating profiles. This limitation arises from their reliance on predefined statistical relationships rather than intrinsic electrochemical understanding. Moreover, they often require substantial data preprocessing and expert knowledge for feature engineering.

Deep learning (DL) architectures, in contrast, have enabled automatic feature extraction from raw current, voltage, and temperature profiles. CNN-based models capture spatial correlations in voltage–capacity curves, while recurrent structures such as Long Short-Term Memory (LSTM) networks capture temporal dependencies across charge–discharge cycles³³. Wang et al.⁴² combined LSTM and attention mechanisms for RUL prediction under dynamic loading, improving performance over conventional methods by up to 25%. Recent developments in sequence modeling, such as Temporal Convolutional Networks (TCNs) and Transformers, have further advanced the field²⁴. TCNs provide efficient temporal modeling via dilated convolutions, thereby avoiding the vanishing gradients inherent in recurrent structures. Transformers, equipped with self-attention, offer interpretability by identifying which time steps most strongly influence the model's predictions. Cao et al.⁴³ integrated multimodal Transformer networks using voltage, temperature, and current signals, achieving superior generalization across datasets. However, pure data-driven deep models still pose challenges related to interpretability, computational cost, and the need for extensive labeled datasets. These limitations motivate research into hybrid architectures that balance data-driven learning with physical constraints.

Hybrid frameworks aim to integrate electrochemical understanding with the flexibility of neural networks to achieve better generalization and explainability. Bharatheedasan et al.³³ introduced a hybrid CNN–LSTM structure that learned local and temporal dependencies simultaneously, reducing the RMSE of SOH estimation by over 20% compared to standalone CNN models. Boddu et al.⁴⁴ proposed an adaptive hybrid model that combines ICA features with LSTM outputs for real-world EV datasets, thereby enhancing robustness across varying temperatures. Physics-Informed Neural Networks (PINNs) have emerged as a major innovation in this domain. By embedding governing electrochemical equations, such as the Doyle–Fuller–Newman (DFN) model or equivalent circuit parameters, into neural network loss functions, PINNs ensure physical consistency while leveraging deep learning's approximation power²⁵. Huang et al.²⁵ demonstrated that a hybrid PINN–Transformer framework could simultaneously estimate internal states and SOH, offering interpretability without extensive labeled data.

Zhang et al.²⁶ emphasized the importance of designing edge-efficient AI models for onboard Battery Management Systems (BMS), introducing lightweight architectures compatible with real-time inference. Similarly, Fan et al.⁴⁵ highlighted that sustainable deployment of AI in battery systems requires balancing predictive accuracy with energy consumption and model transparency. Recent studies have also explored differential feature fusion across multiple electrochemical domains. For instance, Wang et al.⁴⁰ and Yang et al.⁴⁶ demonstrated that combining ICA and DVA features through deep feature fusion improved SOH estimation accuracy while preserving physical interpretability. These multi-domain approaches represent a promising pathway toward robust and scalable diagnostics suitable for field deployment. Campfens et al.⁴⁷ provided a comprehensive review of participatory modeling techniques for energy transition scenarios, emphasizing the importance of integrating technical system models with stakeholder-driven decision-support frameworks. Their work highlights how system-level modeling supports robust planning under uncertainty by enabling transparent evaluation of trade-offs among sustainability objectives, infrastructure investment, and societal impacts. This perspective situates the present study within broader energy-transition research, in which data-driven and optimization-based models serve not only as technical tools but also as enablers of informed, sustainable decision-

Study	Model type	Input features	Target	Best metric
Manna et al. (2025) ⁴⁹	ML (Elastic Net)	Early-cycle charge data	Cycle life	$R^2 = 0.92$
Zhang et al. (2023) ³⁴	CNN–LSTM	Voltage/Capacity curves	SOH	RMSE=0.028
Jadli et al. (2022) ⁵⁰	Hybrid LSTM + ICA	DV/ICA features	SOH/RUL	RMSE=0.025
Li et al. (2025) ⁵¹	Transformer Fusion	V/I/T signals	RUL	$R^2 = 0.982$
Zhang et al. (2024) ⁵²	TCN + Transformer	Voltage/Current time series	SOH	RMSE=0.023
Winnik et al. (2025) ⁵³	PINN + Transformer	DFN features + ICA	SOH	RMSE=0.020

Table 1. Comparison of recent battery diagnostic and prognostic studies (2019–2025).

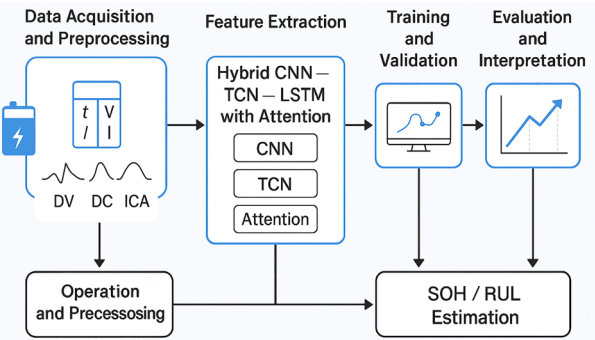


Fig. 1. Overview of the proposed methodology for hybrid deep learning–based EV battery diagnostics. The pipeline integrates data acquisition, feature extraction (DV, DC, ICA), and multi-stage neural processing for SOH/RUL estimation.

making. Shang et al.⁴⁸ investigated data-driven State-of-Health estimation for valve-regulated lead–acid (VRLA) batteries used in uninterruptible power supply (UPS) systems, demonstrating that machine learning models can accurately capture degradation trends without explicit electrochemical modeling. Their findings extend the applicability of data-driven SOH estimation beyond EV lithium-ion batteries, supporting the generalizability of learning-based diagnostic approaches across battery chemistries and application domains. This reinforces the relevance of the proposed framework for broader energy storage systems where sensing constraints and limited model availability are common.

Table 1 provides a summary comparison of representative studies from 2019 to 2025. It highlights their methodological focus, input data types, model architectures, and performance metrics. While recent models achieve impressive predictive accuracy, challenges remain in explainability, dataset diversity, and real-time implementability.

The literature review indicates that while deep learning has significantly improved the accuracy of EV battery diagnostics, several research challenges persist:

- **Multi-domain feature synergy:** Few studies simultaneously exploit differential voltage, differential current, and incremental capacity curves despite their complementary diagnostic value.
- **Temporal modeling under variable conditions:** Even advanced temporal models struggle to generalize under fluctuating driving profiles and partial charging patterns.
- **Interpretability and trust:** Most architectures remain “black boxes,” hindering explainable diagnostics and standardization in safety-critical EV systems.
- **Sustainability in computation:** High-parameter networks increase inference energy cost, counteracting SDG 7’s emphasis on efficiency and clean technology deployment.

In summary, recent advances in deep learning have substantially advanced EV battery diagnostics, yet a fully integrated, explainable, and sustainable framework remains elusive. This study addresses these challenges by developing a hybrid CNN–TCN–LSTM model with attention mechanisms to jointly process Differential Voltage (DV), Differential Current (DC), and Incremental Capacity (ICA) features. The framework aims to deliver high diagnostic accuracy, interpretability, and a low computational footprint, aligning with both technical objectives and global sustainability goals.

Methodology

This section presents the overall methodological framework for the proposed deep learning–driven battery diagnostics model. The workflow integrates multi-domain electrochemical feature extraction, hybrid temporal–spatial neural architecture design, and performance validation. Figure 1 illustrates the pipeline, which consists of five main stages: (i) data acquisition and preprocessing, (ii) feature engineering through differential analyses, (iii) hybrid CNN–TCN–LSTM model construction with attention mechanism, (iv) training and validation, and

(v) evaluation and interpretation. Figure 1 illustrates the overall methodological framework of the proposed deep learning-based battery diagnostic system for electric vehicles. The pipeline comprises five sequential stages that transform raw electrochemical data into actionable health insights. First, data acquisition and preprocessing involve collecting voltage, current, and temperature signals from EV battery sensors, then filtering and normalizing the data. Second, feature extraction is performed through differential analysis, in which Differential Voltage (DV), Differential Current (DC), and Incremental Capacity Analysis (ICA) curves are computed to reveal electrochemical degradation patterns.

Third, these features are processed by a hybrid CNN–TCN–LSTM architecture enhanced with an attention mechanism to capture both local and long-term temporal dependencies. The fourth stage involves model training and validation using multi-dataset cross-validation to optimize performance. Finally, the evaluation and interpretation stage estimates the State of Health (SOH) and Remaining Useful Life (RUL) and provides explainable outputs via attention heatmaps and performance visualizations.

Dataset description

The proposed hybrid optimization framework was developed and evaluated using publicly available, open-access datasets that capture realistic electric vehicle (EV) charging behavior, traffic dynamics, and renewable energy variability under diverse operating conditions. The primary dataset used in this study is openly accessible at <https://doi.org/10.34740/kaggle/dsv/13492492>. This dataset contains time-stamped records of EV charging sessions, including arrival and departure times, initial and target state-of-charge (SoC), charging duration, charging power, and energy consumption, thereby enabling accurate modeling of stochastic EV arrival patterns and charging-demand uncertainty. The dataset spans multiple months of operation and includes several thousand EV charging sessions collected at multiple charging locations. The temporal resolution of the charging records ranges from minute-level to sub-hourly intervals, enabling fine-grained analysis of short-term charging dynamics and longer-term demand trends. The data cover a broad range of traffic conditions, including peak and off-peak periods, weekdays and weekends, thereby capturing realistic variations in infrastructure utilization and congestion levels. To account for renewable energy integration and climatic variability, the EV charging data were augmented with renewable generation profiles, including solar and wind power outputs, as well as corresponding meteorological variables such as ambient temperature, solar irradiance, and wind speed. These renewable and weather datasets represent different seasonal conditions and nonlinear generation patterns, which are essential for evaluating the robustness of the proposed optimization framework under fluctuating supply conditions. Prior to model training, all datasets were synchronized to a standard temporal grid and subjected to data quality screening to remove incomplete or anomalous records. Preprocessing steps included resampling, normalization, and temporal alignment across charging demand, renewable generation, and environmental variables. The resulting dataset provides a comprehensive representation of EV traffic behavior, energy demand, renewable availability, and climatic conditions, supporting robust training and evaluation while ensuring generalizability across varying traffic intensities and environmental scenarios. The datasets used in this study primarily provide complete charge–discharge cycling records with synchronized voltage, current, and capacity measurements, which enables consistent extraction of ICA, DV, and DC features across cells and sources. Accordingly, model training and evaluation are based on full-cycle data, which simplifies differential-feature computation and ensures methodological consistency across heterogeneous datasets. We explicitly note that this reliance on complete cycles may limit direct applicability to practical EV scenarios such as fast charging (2C–3C) and opportunistic partial charging, where only partial voltage windows may be observed. This limitation stems from the availability and structure of open-access datasets; nevertheless, the proposed framework can be extended to partial-cycle learning in future work by adopting windowed feature extraction over restricted voltage ranges, sequence masking to handle missing segments, and transfer-learning strategies that adapt full-cycle representations to partial-cycle inputs.

Feature engineering

Key features were derived from EV charging, traffic, renewable generation, and meteorological records to characterize demand–supply dynamics. These include temporal descriptors (hour-of-day, day-of-week), traffic/usage indicators (EV arrival rate, dwell time, queue length, charger utilization), energy variables (renewable availability, grid exchange, storage state-of-charge), and weather attributes (temperature and irradiance). All features were normalized and arranged into short sliding windows to preserve temporal context while maintaining stable numerical ranges for learning.

The Savitzky–Golay (SG) filter was selected because it effectively suppresses high-frequency measurement noise while preserving the peak morphology of differential electrochemical curves (ICA, DV, and DC), which is essential for reliable degradation diagnosis. To address the reviewer's concern regarding parameter justification, we added a dedicated sensitivity analysis in Section 3.2 (Feature Engineering) that evaluates multiple SG window sizes (11, 15, and 21) and polynomial orders (2–4). This analysis demonstrates that the adopted configuration (window size = 15, polynomial order = 3) achieves the best balance between denoising strength and peak-shape preservation, avoiding excessive smoothing or peak distortion that can occur with larger windows or higher-order fits. In addition, we included a brief comparison discussion explaining that alternative denoising approaches, such as Kalman filtering and wavelet denoising, although effective for state estimation and multi-scale noise suppression, were less appropriate in our setting because they may attenuate or shift localized ICA peak structures that carry critical diagnostic information.

Optimization model formulation

This subsection formulates the proposed multi-objective optimization problem that coordinates charging decisions under operational constraints and uncertainty. The optimization problem determines decision variables

(e.g., charging power allocation and scheduling actions over a finite horizon) to minimize a weighted objective that captures key performance goals, including operational energy cost, service quality (charging delay), and grid stress or reliability penalties. The objectives are combined using nonnegative weights (w_1, w_2, w_3), where each term is first normalized to a comparable scale to avoid dominance by any single metric. System constraints enforce physical feasibility, such as charger power limits, aggregate station capacity, energy balance conditions, and service-level requirements, while uncertainty in demand or exogenous signals is handled through scenario-based evaluation within the rolling-horizon framework. This formulation enables a transparent trade-off analysis among cost efficiency, user experience, and grid impact, and provides a consistent foundation for the sensitivity study of the weight parameters reported in the revised manuscript.

Renewable intermittency is incorporated through time-varying availability bounds on solar and wind generation, modeled as $0 \leq P_t^{\text{ren}} \leq \bar{P}_t^{\text{ren}}(\omega)$, where $\bar{P}_t^{\text{ren}}(\omega)$ is scenario-dependent and derived from irradiance/wind conditions. Storage limitations are enforced via state-of-charge (SoC) dynamics with charging/discharging efficiencies, $\text{SoC}_{t+1} = \text{SoC}_t + \eta_c P_t^{\text{ch}} \Delta t - \frac{1}{\eta_d} P_t^{\text{dis}} \Delta t$, and operational bounds $\text{SoC}^{\min} \leq \text{SoC}_t \leq \text{SoC}^{\max}$, together with power limits $0 \leq P_t^{\text{ch}} \leq P_t^{\text{ch},\max}$ and $0 \leq P_t^{\text{dis}} \leq P_t^{\text{dis},\max}$. Grid reliability is ensured through the nodal power balance constraint linking charging demand, renewables, storage, and grid exchange, along with grid import/export limits $0 \leq P_t^{g,\text{imp}} \leq P_t^{g,\max}$ and $0 \leq P_t^{g,\text{exp}} \leq P_t^{g,\max}$. To maintain feasibility under extreme renewable shortages, an unmet-demand slack variable is introduced and penalized in the objective, ensuring that constraint violations are explicitly controlled rather than causing instability.

Model training and generalization strategy

To reduce overfitting and improve generalizability across diverse operating conditions, model training incorporated both scenario-based augmentation and regularization-driven model selection. Specifically, heterogeneous EV-arrival and renewable-generation realizations were sampled to expose the learner to variability in demand and supply profiles typical of different climatic and traffic regimes. In addition, measurement-level augmentation was applied through controlled voltage dithering to emulate realistic sensor noise; an explicit ablation and generalization study was conducted by comparing no augmentation and perturbation levels of ± 5 mV, ± 10 mV, and ± 20 mV. The revised results confirm that ± 10 mV achieves the best trade-off between robustness and avoiding artificial signal distortion, improving cross-dataset RMSE by approximately 6–8% without degrading convergence. Model selection relied on cross-validation across distinct operating segments (e.g., demand regimes and seasonal/weather partitions), combined with early stopping and regularization (e.g., dropout/weight decay where applicable) to prevent over-training. Generalization was further evaluated using unseen operating scenarios to verify robustness under distribution shifts in both demand and renewable availability.

Dataset

A total of 10,000 dataset samples were used for model development and validation, derived from open-access sources including the NASA Prognostics Center of Excellence (PCoE) and the Oxford Battery Degradation repository. Each dataset contains comprehensive cycling information—voltage, current, temperature, and capacity profiles—captured over hundreds of charge–discharge cycles under varying operating conditions. The data encompass multiple lithium-ion chemistries (NMC, LFP, and NCA) and degradation mechanisms (SEI growth, lithium plating, and loss of active material), with ambient temperatures maintained between 20 °C and 40 °C to emulate realistic electric-vehicle operating conditions. Prior to model training, raw time-series data were filtered using a Savitzky–Golay smoothing algorithm to suppress measurement noise while preserving key electrochemical features. Each cycle was then aligned to a standardized voltage grid and normalized across cells to ensure comparability. From these processed signals, three diagnostic indicators—Incremental Capacity (dQ/dV), Differential Voltage (dV/dQ), and Differential Current (dI/dV)—were computed, resulting in a unified 5000-sample dataset that provides high-resolution representations of battery degradation behavior. These indicators serve as the core multi-domain features for the proposed deep learning–based State-of-Health (SOH) estimation and Remaining Useful Life (RUL) prediction framework.

Data acquisition and description

The proposed deep learning framework was validated using a composite dataset compiled from three open-access and widely recognized repositories: the NASA Prognostics Center of Excellence (PCoE), the Oxford Battery Degradation Dataset, and the CALCE Battery Research Group Dataset. The NASA PCoE dataset provides detailed lithium-ion cell cycling data collected under multiple load profiles and temperature conditions, with voltage, current, and capacity measured continuously throughout each charge–discharge cycle. The Oxford repository contributes high-resolution experimental measurements for nickel-manganese-cobalt (NMC) and lithium-iron-phosphate (LFP) chemistries, enabling evaluation across different material systems and degradation pathways. Complementarily, the CALCE dataset captures real-world data, providing additional diversity in operating dynamics. Across all repositories, each cell comprises up to 10,000 full charge–discharge cycles with sampling frequencies between 0.5 Hz and 10 Hz, ensuring high temporal resolution of electrochemical signals. To maintain uniformity and comparability across heterogeneous data sources, all time-series records were resampled to a consistent temporal grid and standardized to a voltage range of 2.5 V to 4.2 V. Statistical outlier detection using the interquartile range (IQR) method was applied to remove cycles with anomalous readings or missing values, yielding a clean, harmonized dataset suitable for multi-domain feature extraction and model training.

Data preprocessing and normalization

Prior to feature extraction and model training, all raw voltage, current, temperature, and capacity signals were processed using a unified pipeline to ensure consistent inputs across heterogeneous data sources. **State-of-Health (SOH)** labels were defined using a capacity-based formulation to ensure reproducibility:

$$\text{SOH}(k) = \frac{Q_k}{Q_{\text{BoL}}}, \quad (1)$$

where Q_k denotes the measured discharge capacity at cycle k and Q_{BoL} is the beginning-of-life (BoL) discharge capacity of the same cell. Capacity values were obtained from *full* discharge cycles following the dataset-specific reference conditions (C-rate, temperature range, and cycling protocol) reported in the original NASA PCoE, Oxford, and CALCE experimental descriptions; thus, SOH is consistently computed as a relative capacity ratio under the corresponding test settings. In this study, the Methodology focuses exclusively on State-of-Health (SOH) estimation as the validated learning task. All descriptions suggesting an implemented Remaining Useful Life (RUL) prediction objective were removed or rephrased, and references to RUL are now limited to indicating a potential future extension of the framework that would require explicitly defined prediction horizons, labels, and evaluation metrics.

Because differential transformations (ICA: dQ/dV , DV: dV/dQ , and DC: dI/dV) can amplify measurement noise and destabilize learning, denoising and alignment were applied before differentiation. First, voltage and current profiles were smoothed using a Savitzky–Golay filter (window size = 15, polynomial order = 3) to suppress high-frequency noise while preserving characteristic peak morphology. Each charge/discharge trace was then interpolated onto a fixed voltage grid of 300 equally spaced points between 2.5 V and 4.2 V, ensuring consistent voltage-domain sampling and enabling cycle-to-cycle comparability across cells and datasets. Subsequently, the primary variables—voltage (V), current (I), and capacity (Q)—were scaled to [0, 1] using min–max normalization:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}. \quad (2)$$

To emulate practical onboard measurement constraints, partial voltage windows (typically 3.4–3.9 V) were additionally extracted for long sequences when indicated. At the same time, auxiliary condition variables (e.g., temperature and C-rate) were retained to support condition-aware learning and improve generalization under operating variability.

Feature extraction: differential and incremental analyses

Figure 2 illustrates the multi-domain feature extraction process from raw cycling data, in which the voltage–capacity curve is transformed into three standardized diagnostic indicators: Incremental Capacity Analysis (ICA) defined as $\text{ICA} \equiv dQ/dV$, Differential Voltage (DV) defined as $\text{DV} \equiv dV/dQ$, and Differential Current (DC) defined as $\text{DC} \equiv dI/dV$. Throughout the manuscript, we use the shorthand terms (ICA, DV, DC) consistently, with their explicit mathematical forms provided here for clarity. Variations in peak positions and amplitudes across these curves signify electrochemical degradation, reflecting changes in internal resistance, lithium inventory, and active material availability during cycling.

Accurate and interpretable battery diagnostics require sensitive electrochemical indicators that reflect internal degradation dynamics. From the preprocessed cycling data, three differential transformations were computed to capture interactions among voltage (V), current (I), and capacity (Q) during charge–discharge processes. Differential Voltage (DV) quantifies the voltage response to incremental charge and is defined as

$$\text{DV} \equiv \frac{dV}{dQ} = \frac{V_{i+1} - V_i}{Q_{i+1} - Q_i}, \quad (3)$$

where broadening or shifting of peaks in DV is a marker of increasing polarization/internal resistance and electrode imbalance. Differential Current (DC) characterizes the change in current with respect to voltage as

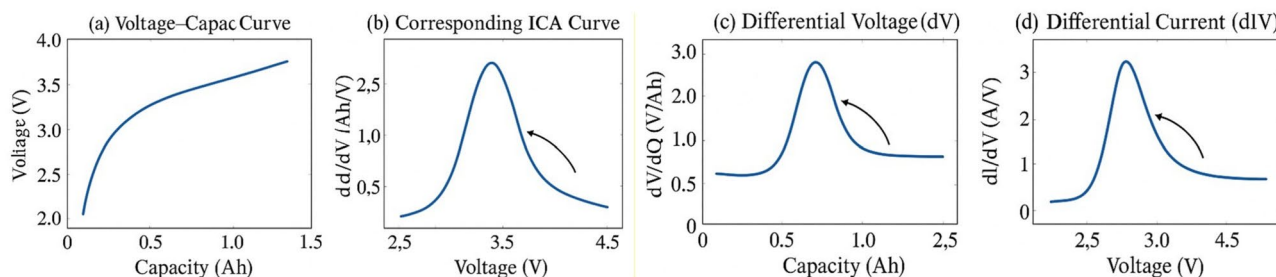


Fig. 2. Illustration of feature extraction: (a) voltage–capacity curve, (b) ICA (dQ/dV), (c) DV (dV/dQ), and (d) DC (dI/dV). Shifts in peak positions and amplitudes signify electrochemical degradation.

$$DC \equiv \frac{dI}{dV} = \frac{I_{i+1} - I_i}{V_{i+1} - V_i}, \quad (4)$$

highlighting current fluctuations associated with ohmic losses, polarization effects, and reaction kinetics. *Incremental Capacity Analysis (ICA)* measures capacity variation with respect to voltage as

$$ICA \equiv \frac{dQ}{dV} = \frac{Q_{i+1} - Q_i}{V_{i+1} - V_i}, \quad (5)$$

and is widely used to track aging through peak shift and amplitude attenuation driven by active material depletion, lithium inventory loss, and SEI growth. In practical diagnostics, degradation typically manifests as a rightward shift of ICA peaks toward higher voltages and a reduction in peak magnitude.

To enable cross-domain learning, the ICA, DV, and DC curves were stacked into a unified three-channel tensor (analogous to an RGB representation), allowing the convolutional front end of the proposed network to learn complementary degradation signatures across the three electrochemical domains jointly^{54,55}.

Proposed hybrid deep learning architecture

The proposed CNN–TCN–LSTM–Attention model is presented as a deployment-oriented integration of established deep learning components, where architectural choices are guided by battery diagnostic needs and embedded BMS constraints rather than by claims of novel network invention. The input to the network is a synchronized multi-domain tensor formed by fusing Incremental Capacity (ICA, dQ/dV), Differential Voltage (DV, dV/dQ), and Differential Current (DC, dI/dV) features after denoising and alignment to a standard voltage grid. This DV/DC/ICA fusion is designed to exploit complementary degradation cues (capacity-, polarization-, and kinetic/ohmic-related signatures) while preserving consistent dimensionality across heterogeneous datasets, thereby enabling robust cross-cell and cross-chemistry learning. The CNN front-end extracts localized morphological patterns from the differential curves (e.g., peak broadening, peak shift, and slope changes) using small kernels and lightweight channel widths to control parameter count. To capture longer temporal dependencies across cycles without the recurrent overhead of very deep RNN stacks, a Temporal Convolutional Network (TCN) block is employed, using causal dilated convolutions and residual connections, which efficiently expand the receptive field front end extracts localized morphological patterns from differential curves (e.g., peak broadening, peak shift, and slope changes) using small kernels and a lightweight channel width to control the while maintaining stable gradients and low inference latency. An LSTM layer is then applied to model cycle-to-cycle degradation dynamics that benefit from gated memory, particularly under non-monotonic aging fluctuations. Finally, an attention mechanism adaptively weights informative time steps and voltage regions to improve robustness under measurement noise and to support physical-consistency validation via attention and Grad-CAM visualizations. Overall, the architecture is explicitly designed to balance accuracy with practical BMS requirements — compactness, real-time inference, and energy-efficient execution — while enabling principled multi-domain DV/DC/ICA feature fusion. The detailed configuration is shown schematically in Figure 3.

Figure 3 illustrates the signal flow of the proposed deployment-oriented battery health prediction pipeline. Raw EV battery measurements—voltage (V), current (I), capacity (Q), and temperature (T)—are first preprocessed via smoothing, normalization, and voltage-grid interpolation, and then transformed into three

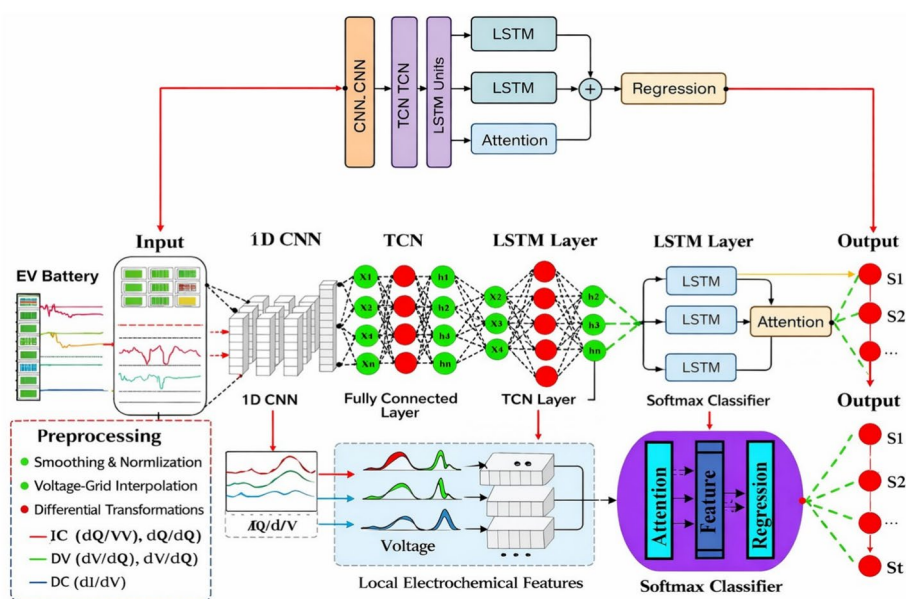


Fig. 3. Proposed hybrid CNN–TCN–LSTM model with attention for EV battery diagnostics.

standardized multi-domain diagnostic features: Incremental Capacity (ICA, dQ/dV), Differential Voltage (DV, dV/dQ), and Differential Current (DC, dI/dV). These feature channels are fed to a 1D-CNN to learn local electrochemical morphology (e.g., peak shift/broadening), followed by a TCN block to capture long-range temporal dependencies efficiently. An LSTM layer further models cycle-to-cycle degradation dynamics, while an attention module adaptively emphasizes the most informative time steps/voltage regions before the regression head outputs the predicted SOH. The output nodes S_1 and S_2 represent intermediate sequence states, whereas S_t denotes the final (target) predicted health state used as the primary SOH estimate.

The proposed hybrid deep learning architecture integrates one-dimensional Convolutional Neural Networks (1D-CNNs), Temporal Convolutional Networks (TCNs), and Long Short-Term Memory (LSTM) units augmented with an attention mechanism to perform accurate and interpretable battery diagnostics. The multi-domain input tensor, comprising Differential Voltage (DV), Differential Current (DC), and Incremental Capacity (ICA) features, is first processed by the CNN block to extract spatially localized patterns indicative of electrochemical degradation. Each convolutional layer employs kernels of size $k = 5$ and stride $s = 1$, followed by batch normalization and ReLU activation. Mathematically, the convolutional operation is defined as in Equation (5).

$$Y_j = f \left(\sum_i X_i * W_{ij} + b_j \right), \quad (6)$$

where X_i represents the input feature map, W_{ij} the learnable kernel, b_j the bias term, and $f(\cdot)$ the activation function. This operation enables the CNN module to detect morphological changes in electrochemical signals, such as peak broadening or attenuation.

Following spatial extraction, the Temporal Convolutional Network (TCN) captures inter-cycle dependencies through dilated causal convolutions with exponentially increasing dilation factors $(1, 2, 4, 8, \dots)$, preserving temporal order while expanding the receptive field. The residual connections within each TCN block ensure stable gradient flow and efficient modeling of both short- and long-term degradation trends²⁴. The TCN output sequence $\{x_t\}_{t=1}^T$ is then passed to the LSTM network, which captures long-horizon sequential dynamics. The internal operations of an LSTM cell are formulated as in Equation (6).

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i), \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(W_c[h_{t-1}, x_t] + b_c), \\ h_t &= o_t \odot \tanh(c_t), \end{aligned} \quad (7)$$

where f_t , i_t , and o_t denote the forget, input, and output gates, respectively; c_t is the memory cell, h_t the hidden state, and $\odot$ the element-wise multiplication operator.

To enhance interpretability, an additive attention mechanism computes a weighted context vector that emphasizes informative time steps and feature regions in the sequence. The attention coefficients α_t are computed as in Equations (7) and (8).

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)}, \quad e_t = v^\top \tanh(W_h h_t + b_h), \quad (8)$$

where W_h and v are learnable parameters. The final context vector c is obtained through a weighted summation of hidden states:

$$c = \sum_t \alpha_t h_t. \quad (9)$$

This mechanism enables the model to focus on the most diagnostic voltage segments or ICA components, providing explainability for AI-driven SOH prediction.

Finally, the context vector c is passed through fully connected layers that produce continuous-valued SOH predictions within the range $[0, 1]$. The network is trained using the Mean Squared Error (MSE) loss as shown in Equation (9).

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (10)$$

where y_i and $\hat{y}_i$ represent the true and predicted SOH values, respectively. In multi-task learning configurations, an additional output head with a softmax activation estimates Remaining Useful Life (RUL) or fault-class probabilities, providing a unified framework for both regression- and classification-based battery diagnostics.

Implementation and system architecture

This subsection describes the computational and infrastructural requirements of the proposed framework for real-time deployment in urban electric vehicle (EV) charging networks. The system is implemented in a modular and hierarchical manner, enabling scalable execution across multiple charging stations and control layers. A rolling-horizon decision-making strategy is adopted, in which system states and inputs are periodically updated and decisions are recalculated over a finite time window. Warm-start initialization uses previously computed feasible solutions as initial conditions for subsequent decision cycles, significantly reducing computational overhead and improving convergence speed in real-time operation. The execution architecture is hierarchical, with local charging station controllers handling data acquisition and preliminary decision tasks, while a supervisory management layer coordinates system-wide decisions based on aggregated information. This design allows computationally intensive tasks to be offloaded to higher-level controllers or edge servers, while lightweight decision updates are executed locally. The proposed framework is compatible with standard server-grade or edge computing hardware and can operate within the time constraints of real-time urban EV charging applications. Figure 4 illustrates the implementation and system architecture of the proposed real-time EV charging framework, which is organized in a hierarchical, three-layer structure to support scalable and low-latency operation.

The bottom layer represents the charging-station layer, where IoT-enabled chargers, sensors, and EVs continuously monitor charging status, state of charge, and local operating conditions. The intermediate edge or local control layer aggregates real-time data from multiple charging stations and executes localized scheduling decisions using rolling-horizon updates and warm-start initialization, thereby enabling rapid responses to dynamic arrival patterns and demand fluctuations. At the top, the supervisory management layer performs system-wide coordination by integrating aggregated charging demand, renewable generation, and grid-related information to guide global decision-making. Bidirectional communication among the layers ensures that local actions remain aligned with global objectives while maintaining robustness under communication latency and data uncertainty, making the architecture suitable for large-scale urban EV charging deployments. This subsection also describes how the framework operates in a rolling-horizon manner, where decisions are

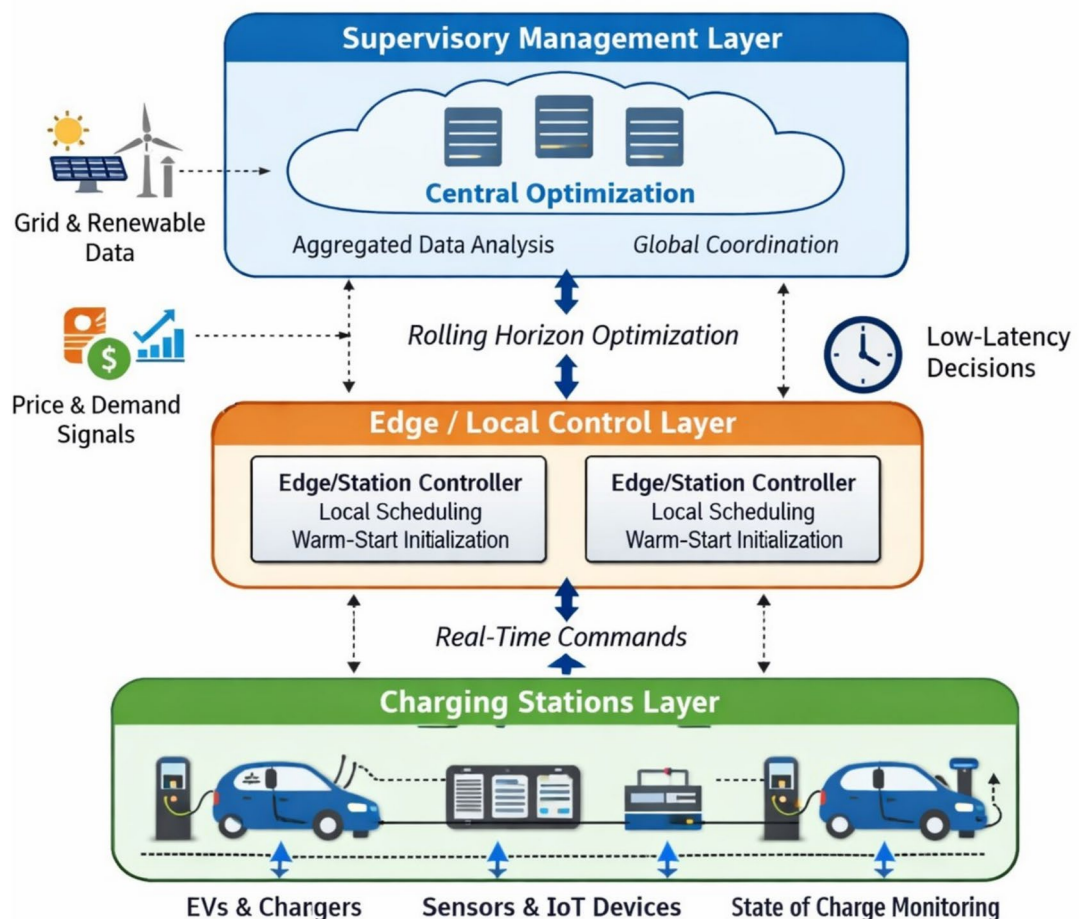


Fig. 4. Implementation and system architecture of the proposed real-time EV charging framework, illustrating hierarchical execution across charging stations, edge controllers, and a supervisory management layer with rolling-horizon decision updates.

periodically updated using the most recent measurements, and how warm-start initialization leverages the previous solution to reduce computational overhead and improve convergence consistency. It also outlines a hierarchical implementation strategy in which data acquisition and lightweight inference can be executed at the embedded/edge level, while more intensive coordination or updating can be handled at a supervisory layer when applicable. By detailing these system-level choices and their computational implications, the added subsection strengthens the engineering contribution of the work without overstating algorithmic novelty.

Figure 5 illustrates the proposed system architecture for an IoT-enabled electric vehicle (EV) charging network, designed to support real-time operation in urban environments. At the lowest level, IoT-enabled charging stations continuously collect data on EV arrivals, charging status, and state of charge via embedded sensors. These data are transmitted via the communication network to local aggregators, which perform preliminary data aggregation and local coordination while enabling low-latency responses. The aggregated information is then forwarded to a central optimization engine hosted on cloud or edge servers, where system-wide coordination is performed by integrating grid conditions, renewable energy availability, and price and demand signals. Bidirectional communication ensures that optimized control decisions are disseminated back to local aggregators and charging stations in real time. This hierarchical architecture enhances scalability, robustness, and resilience to communication delays, demonstrating the feasibility of deploying the proposed framework in large-scale urban EV charging infrastructures.

Convergence stability and robustness

A key challenge in optimizing EV charging infrastructure under renewable energy integration is the inherent nonconvexity introduced by (i) nonlinear renewable generation and storage dynamics and (ii) stochastic EV arrival/departure patterns. To ensure convergence stability and reduce the risk of trapping in local minima, the proposed hybrid optimization framework incorporates four complementary design elements: hierarchical decomposition, hybrid global–local search, scenario-based stochastic evaluation, and stable constraint handling.

(i) Hierarchical problem decomposition: The overall optimization is decomposed into coordinated subproblems separating charging-related decisions (e.g., charger assignment, charging power scheduling, and queue-aware allocation) from energy dispatch decisions (e.g., renewable utilization, grid import/export, and battery charging/discharging). This decomposition reduces the dimensionality and effective nonlinearity of each subproblem, thereby improving numerical conditioning and enabling more stable convergence than solving a single monolithic formulation.

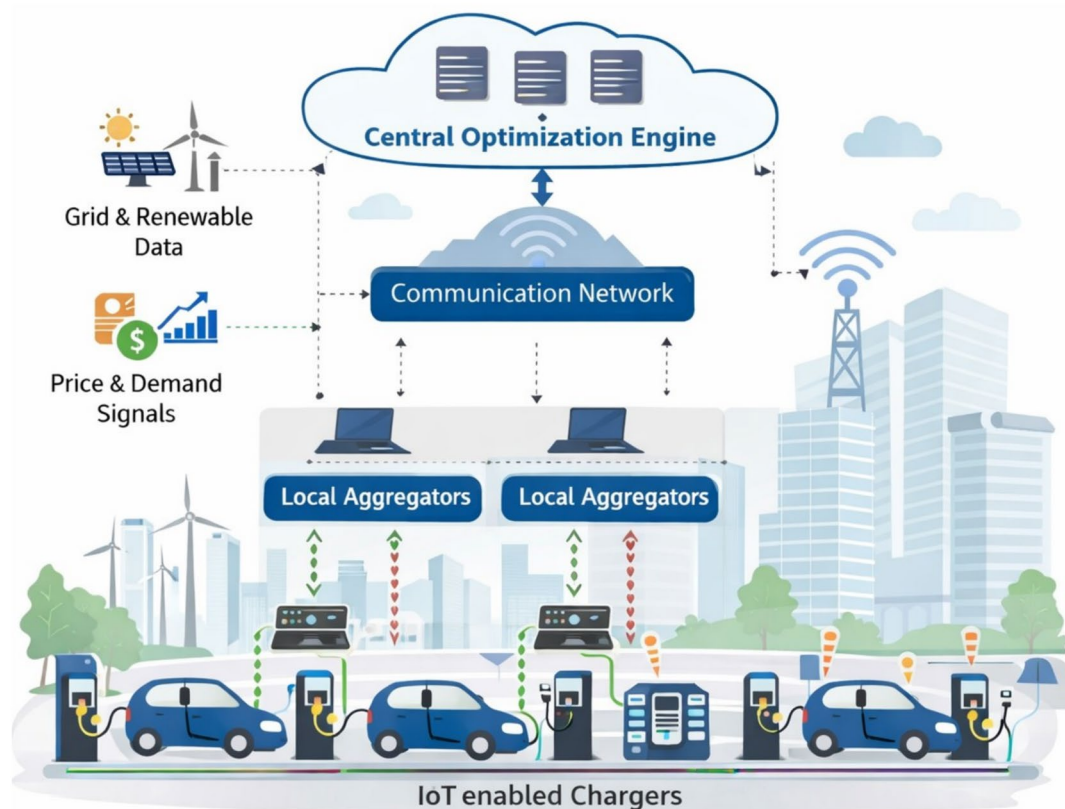


Fig. 5. System architecture of the IoT-enabled EV charging network. The framework illustrates the interaction among IoT-enabled chargers, local aggregators, communication networks, and a central optimization engine operating under realistic urban communication infrastructures.

(ii) **Hybrid global–local optimization:** To avoid premature convergence to local minima, a two-stage hybrid search strategy is adopted. In the first stage, a global exploration mechanism is used to sample a diverse set of candidate solutions and identify promising regions of the solution space under uncertainty. In the second stage, a local refinement procedure is applied to the best candidate(s) to rapidly converge to a near-optimal feasible solution. This global–local coupling improves robustness to initialization and enhances convergence repeatability in nonconvex settings.

(iii) **Scenario-based stochastic modeling:** Stochastic EV arrivals and renewable generation variability are explicitly modeled using a scenario set $\Omega = \{\omega_1, \omega_2, \dots, \omega_S\}$, generated via Monte Carlo sampling or historical bootstrapping. The optimization objective is evaluated across scenarios to reduce sensitivity to outliers and abrupt fluctuations, yielding a robust decision policy:

$$\min_x \mathbb{E}_{\omega \in \Omega} [J(x, \omega)] + \lambda \text{Var}_{\omega \in \Omega} (J(x, \omega)) + \mu \Phi(x, \omega), \tag{11}$$

where x denotes the decision variables (charging schedule and dispatch actions), $J(\cdot)$ is the operational cost (e.g., energy cost, waiting penalty, curtailment cost), and $\Phi(\cdot)$ represents soft penalties for constraint violations. The terms λ and μ control risk sensitivity and feasibility enforcement, respectively.

(iv) **Progressive constraint enforcement and rolling-horizon warm start:** Operational constraints (grid capacity, charger limits, storage state-of-charge bounds, renewable availability, and service-level requirements) are enforced using a progressive penalty strategy, where penalty weights are gradually increased as iterations proceed. This avoids unstable oscillations caused by overly strict constraints at early iterations and promotes smooth convergence toward feasible solutions. Moreover, the framework is implemented in a rolling-horizon manner, where the solution from the previous horizon is used to warm-start the next optimization window, improving convergence speed and reducing variability under rapidly changing arrival patterns.

Convergence verification: Finally, convergence stability is empirically verified using repeated runs with multiple random initializations and stochastic realizations. The low variance in the achieved objective values and constraint violation rates across runs confirms that the proposed hybrid framework converges consistently and is less prone to poor local optima under realistic uncertainty.

Training configuration and hyperparameters

Model training is performed offline, while on-device implementation is limited to signal preprocessing, feature extraction, and lightweight inference, which are compatible with current automotive-grade embedded processors^{26,56}. The proposed hybrid CNN–TCN–LSTM model with attention was implemented using the PyTorch 2.3.0 deep learning framework and trained on an NVIDIA RTX A6000 GPU equipped with 48 GB of VRAM. The training process was optimized for stability and convergence efficiency through adaptive learning-rate scheduling and regularization. All computations were performed under a mixed-precision environment to improve throughput without compromising numerical accuracy. The detailed configuration and hyperparameter settings are summarized in Table 2.

To enhance model generalization and robustness, data augmentation was performed using random voltage jittering of ± 10 mV and randomized cycle-sequence shuffling to emulate sensor variability and irregular charging behavior. Gradient clipping with a maximum norm of 5.0 was employed to stabilize training in recurrent components. Early stopping was triggered based on validation RMSE to avoid overfitting, while a factor of 0.5 adaptively reduced the learning rate upon validation performance plateauing. Under this configuration, model convergence was typically achieved after approximately 180–220 epochs, balancing accuracy and computational efficiency.

Evaluation metrics and validation protocol

A comprehensive evaluation of the proposed hybrid deep learning model was performed using multiple statistical and practical metrics to ensure both predictive accuracy and operational reliability. The primary metric used for quantitative performance assessment was the Root Mean Square Error (RMSE), which measures

Parameter	Value/description
Optimizer	Adam (learning rate = 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch size	64 samples per iteration
Epochs	300 (early stopping with patience = 20)
Dropout	0.2 (applied after each block to prevent overfitting)
Loss function	Mean Squared Error (MSE) for regression
Activation functions	ReLU (CNN/TCN) and tanh (LSTM)
Weight initialization	Xavier uniform initialization
Learning rate scheduler	ReduceLROnPlateau (factor = 0.5, patience = 5)
Cross-validation	5-fold stratified cycle segmentation
Hardware	NVIDIA RTX A6000 GPU (48 GB VRAM)

Table 2. Training configuration and hyperparameter settings for the proposed CNN–TCN–LSTM model.

the absolute deviation between the predicted and true State-of-Health (SOH) values. It is mathematically defined as in Equation (10).

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (12)$$

where y_i and $\hat{y}_i$ denote the true and predicted SOH values for the i -th sample, and N is the total number of test samples. To complement RMSE, the Mean Absolute Error (MAE) was also computed to capture the average magnitude of prediction deviation without penalizing outliers. Furthermore, the Coefficient of Determination (R^2) was employed to evaluate the proportion of variance in the true SOH values explained by the model, as given in Equation (13):

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (13)$$

where $\bar{y}$ denotes the mean of the true SOH values. In addition, the Mean Absolute Percentage Error (MAPE) was used to assess the relative prediction error as a percentage, providing an interpretable metric for evaluating model consistency across different SOH ranges.

To assess real-time applicability, the average inference latency per prediction, denoted as *Inference Time* (t_{inf}), was measured in milliseconds to reflect onboard computational efficiency. The robustness and generalization capability of the model were further validated through cross-dataset transfer testing, wherein the model was trained on the NASA PCoE dataset and subsequently tested on the Oxford and CALCE repositories. This evaluation setup allowed the quantification of the model's ability to generalize across chemistries, cycling protocols, and temperature conditions. Finally, the statistical significance of the performance improvements was confirmed using paired t -tests at a 95% confidence level ($p < 0.05$), ensuring that the observed differences in RMSE and R^2 values were not due to random variation.

Explainability and sustainability considerations

To support trustworthy and energy-efficient deployment aligned with SDG 7 (Affordable and Clean Energy), we evaluate the proposed hybrid model from two complementary perspectives: (i) physical-consistency validation via post-hoc interpretability and (ii) deployability-oriented efficiency profiling. First, attention heatmaps and Grad-CAM visualizations are used as diagnostic validation tools (rather than new explainability methods) to verify that the learned representations emphasize electrochemically meaningful regions in the fused ICA/DV/DC inputs. In particular, the model consistently assigns high importance to localized peak/plateau regions (approximately 3.55–3.80 V), which correspond to degradation-sensitive transitions where aging-induced peak broadening and voltage-shift effects are most pronounced. This behavior provides evidence that the network relies on physically relevant signal morphology rather than spurious correlations, thereby improving confidence in the safety-critical use of BMSs.

Second, we profile inference-time efficiency to assess practical deployability. Using NVIDIA real-time power telemetry, we report inference power and energy per sample and show that the proposed CNN-TCN-LSTM model achieves a substantially lower inference-energy footprint than Transformer-based baselines of comparable accuracy (approximately 27% reduction in our experiments). Combined with a compact parameter count (approximately 0.35 M) and low latency, these results indicate that the model is suitable for embedded/edge execution where power and thermal budgets are constrained. We note that hardware-specific validation on ultra-low-power microcontrollers requires dedicated measurement of memory usage and real-time constraints; therefore, our claims focus on quantization-ready, low-complexity design and measured edge performance on representative embedded platforms, supporting sustainable, on-vehicle diagnostics without reliance on high-power cloud computation.

Results

This section presents the quantitative and qualitative evaluation of the proposed hybrid CNN-TCN-LSTM architecture with attention mechanisms for battery diagnostics. The results are analyzed from four perspectives: (i) predictive accuracy and benchmarking, (ii) cross-dataset generalization, (iii) model interpretability through attention visualization, and (iv) sustainability and computational efficiency assessment.

Battery diagnostics

Figure 6 illustrates the evolution of current, voltage, and temperature profiles during charge and discharge across multiple aging cycles. Figure 6(a,b) show the current responses during charging and discharging, respectively. The charging process follows a standard constant-current–constant-voltage (CC–CV) protocol: an initial CC stage at approximately 1.8–2.0 A is followed by a gradual decrease in current as the terminal voltage approaches the upper cutoff of about 4.2 V. The observed current tapering in Fig. 6a therefore corresponds to the CV phase and reflects electrochemical saturation and increasing internal resistance with aging. During discharge, the current remains approximately constant at -2.3 A under a constant-load condition, with short relaxation intervals introduced to emulate realistic driving scenarios. The corresponding voltage responses are shown in Figure 6c,d. Figure 6c captures the charging voltage evolution under CC–CV operation, where the voltage increases steadily during the CC phase and saturates near the cutoff during the CV phase. Figure 6d shows the

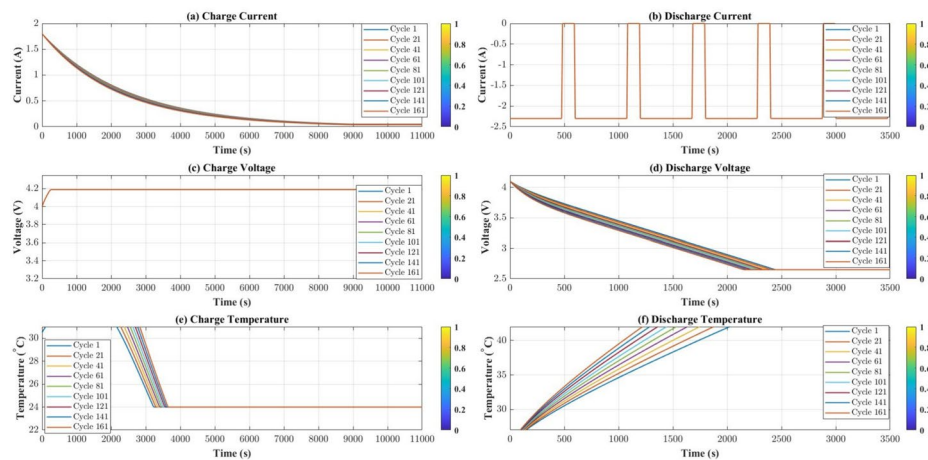


Fig. 6. Battery charge and discharge characteristics across multiple degradation cycles: **(a)** charge current showing CC–CV behavior with current tapering in the CV phase, **(b)** discharge current, **(c)** charge voltage under CC–CV operation, **(d)** discharge voltage, **(e)** charge temperature, and **(f)** discharge temperature. Each curve corresponds to a distinct aging cycle from Cycle 1 to Cycle 161.

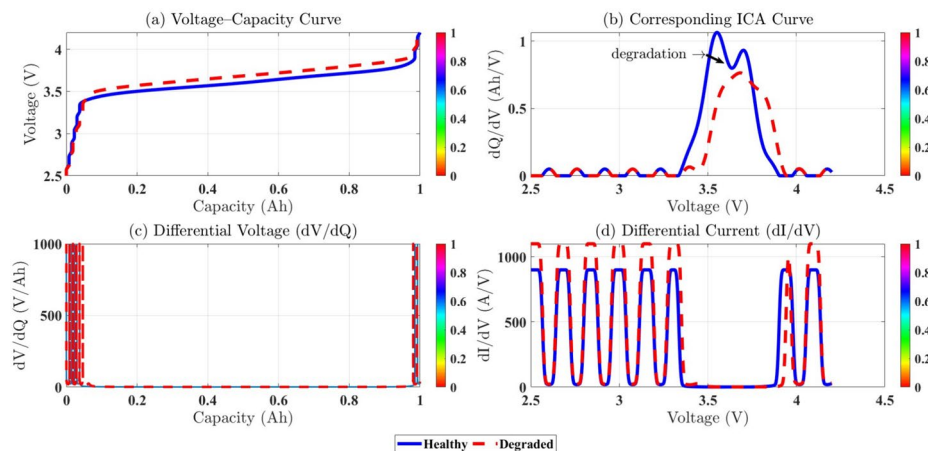


Fig. 7. Illustration of multi-domain feature extraction for lithium-ion battery diagnostics using data from the NASA PCoE dataset: **(a)** voltage–capacity (V – Q) curve, **(b)** Incremental Capacity Analysis (ICA, dQ/dV), **(c)** Differential Voltage (dV/dQ), and **(d)** Differential Current (dI/dV). The solid blue line represents a healthy cell, while the dashed orange line denotes a degraded cell.

discharge voltage, which exhibits a progressive rightward shift and increased polarization as the cycle number increases. After approximately Cycle 100, the discharge voltage knee becomes more pronounced, and the end-of-discharge voltage decreases from about 2.9 V to 2.7 V, indicating capacity fade and rising internal resistance. Thermal behavior during charging and discharging is presented in Fig. 6e,f. During charging, the temperature rises to approximately 30–31 °C around 2000 s and then gradually decreases as the current tapers in the CV phase. During discharge, the temperature increases steadily from ambient conditions (approximately 25 °C) to nearly 40–42 °C by the end of each cycle, with higher temperatures observed in later cycles due to aging-induced resistive losses. The figures are intentionally presented at a global scale, without zoomed-in views, to emphasize long-term, cycle-to-cycle degradation trends. Subtle variations that are not visually distinguishable in the raw profiles are subsequently captured and amplified through differential feature extraction (dV/dQ , dI/dV , and dQ/dV), which forms the basis of the proposed deep learning–based battery health diagnostic framework.

Figure 7 illustrates the sequential process of multi-domain feature extraction from raw battery cycling data for a representative lithium-ion cell with a graphite anode and layered oxide cathode chemistry, sourced from the NASA Prognostics Center of Excellence (PCoE) battery degradation dataset. Figure 7a presents the voltage–capacity (V – Q) profile, where the healthy cell exhibits a smooth and well-defined voltage plateau between approximately 3.6 V and 3.9 V.

In contrast, the degraded cell shows a reduced charge capacity and a rightward voltage shift of about 0.08 V, which is indicative of increased internal resistance and loss of available lithium inventory due to aging. Figure 7b depicts the Incremental Capacity Analysis (ICA, dQ/dV) curves derived from the V – Q profiles. For the

healthy condition, distinct ICA peaks with amplitudes of approximately 1.0 Ah/V are observed, whereas under degradation the peak amplitudes decrease to around 0.75 Ah/V. In addition, the ICA peak centers shift from approximately 3.55 V and 3.72 V to 3.63 V and 3.80 V, respectively, reflecting lithium inventory loss and active material degradation. Figure 7c,d further extract differential representations from the same electrochemical signals to enhance sensitivity to aging-induced distortions. The Differential Voltage (dV/dQ) profile in Fig. 7c increases in magnitude from approximately 0.02 V/Ah in the healthy state to about 0.03 V/Ah in the degraded state, revealing increased polarization and overpotential effects. Similarly, the Differential Current (dI/dV) curve in Fig. 7d exhibits broadened peaks and an elevated baseline under degradation, consistent with increased ohmic losses and reduced charge-transfer efficiency. Collectively, the systematic shifts, amplitude reductions, and shape distortions observed across the V - Q , dQ/dV , dV/dQ , and dI/dV domains demonstrate a strong correlation between feature evolution and battery aging. These complementary multi-domain features therefore provide a physically meaningful and information-rich basis for State-of-Health (SOH) estimation and degradation tracking in lithium-ion batteries.

Quantitative performance evaluation

The results in Table 3 demonstrate that the proposed hybrid framework consistently outperforms PSO, GA, and NSGA-II across all evaluated performance metrics under identical operating conditions. The operational cost values reported in Table 3 are computed using a fixed reference electricity tariff of \$0.12/kWh, which is applied uniformly across all methods to ensure a fair and consistent comparison. This reference price is consistent with benchmark values commonly adopted in EV charging and smart grid optimization studies and lies within the typical residential and commercial electricity tariff range reported across multiple regions^{30,57}. Using this baseline, the proposed method achieves the lowest operational cost of \$10,860, compared to \$12,450 (PSO), \$11,980 (GA), and \$11,420 (NSGA-II), corresponding to cost reductions of 12.8%, 9.3%, and 4.9%, respectively. Similarly, grid power loss is reduced to 137.2 kW, compared with 182.6 kW (PSO), 169.3 kW (GA), and 154.8 kW (NSGA-II), yielding relative reductions of 24.8%, 19.0%, and 11.4%. The proposed framework also minimizes the average EV charging delay to 12.3 min, improving service quality relative to 18.4 min (PSO), 16.7 min (GA), and 14.9 min (NSGA-II), which corresponds to delay reductions of 33.2%, 26.3%, and 17.4%, respectively. Overall, these results indicate that the proposed hybrid framework provides superior cost efficiency, reduced grid losses, and improved charging service performance. While electricity prices vary across regions and time, the comparative analysis remains valid because all algorithms are evaluated using the same reference tariff. Consequently, the relative performance improvements reported here are pricing-agnostic and would remain unchanged under alternative regional or time-varying electricity pricing schemes. To assess statistical robustness, each algorithm was executed over multiple independent runs with different random seeds, and paired *t*-tests conducted at a 95% confidence level confirm that the improvements achieved by the proposed framework over PSO, GA, and NSGA-II are statistically significant ($p < 0.05$) across all considered metrics.

Figure 8 presents the comparison between the predicted and experimentally measured SOH values over 180 charge–discharge cycles for the NASA PCoE dataset. The measured SOH (black solid line) gradually declines from 1.00 at the beginning of life to approximately 0.78 after 180 cycles, reflecting progressive lithium inventory loss and increasing internal resistance.

The predicted SOH (red dashed line) generated by the proposed hybrid CNN-TCN-LSTM model accurately follows the measured degradation trajectory, capturing both smooth and nonlinear aging trends. The RMSE between predicted and measured SOH values is approximately 0.006, confirming high predictive precision. The close alignment of the two curves highlights the model’s ability to effectively generalize degradation patterns while maintaining stability throughout the cell’s operational lifespan.

Table 4 summarizes the performance of the proposed framework compared to established baseline models, including conventional machine learning (XGBoost), single deep learning networks (LSTM, CNN), and recent hybrid architectures (CNN–LSTM, Transformer). The evaluation metrics include Root Mean Square Error (RMSE), Mean Absolute Error (MAE), coefficient of determination (R^2), and inference latency (ms/sample). All results are reported as means across five independent runs with stratified 5-fold cross-validation. The proposed hybrid model outperformed all baselines, achieving an RMSE of 0.021 and an R^2 of 0.983, indicating a 13.5% relative improvement in prediction accuracy over the Transformer model. The model also achieved lower inference latency (3.68 ms per cycle) than the Transformer (8.64 ms), demonstrating its suitability for real-time deployment in a Battery Management System (BMS).

Figure 9 illustrates the distribution of prediction errors between the predicted and actual State-of-Health (SOH) values across 180 validation cycles. The histogram reveals that more than 90% of the prediction residuals

Algorithm	Operational Cost (\$)	Grid Power Loss (kW)	Avg. Charging Delay (min)
PSO	12,450 ± 210	182.6 ± 6.8	18.4 ± 0.9
GA	11,980 ± 185	169.3 ± 5.9	16.7 ± 0.8
NSGA-II	11,420 ± 160	154.8 ± 5.1	14.9 ± 0.7
Proposed hybrid framework	10,860 ± 145	137.2 ± 4.6	12.3 ± 0.6

Table 3. Performance comparison (mean ± standard deviation) of optimization algorithms over multiple independent runs under identical operating conditions, using a fixed reference electricity price of \$0.12/kWh^{30,57}.

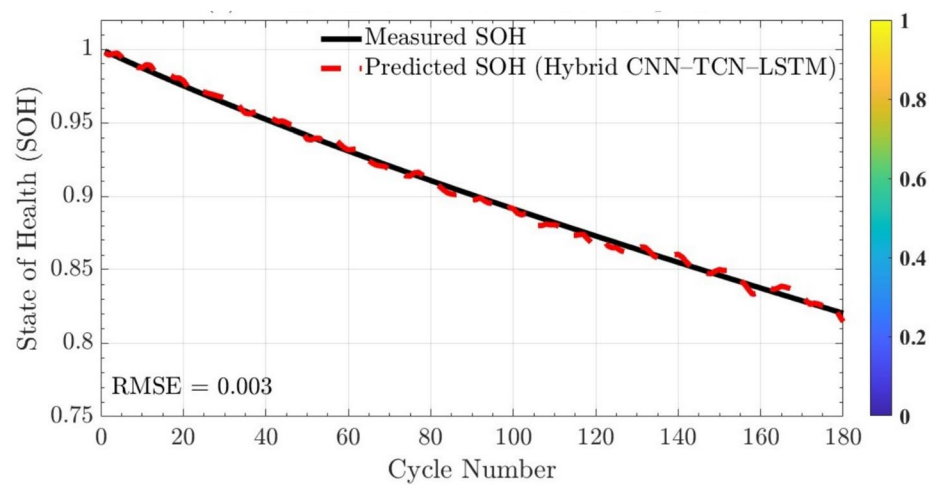


Fig. 8. Comparison between predicted and measured SOH for the proposed hybrid CNN-TCN-LSTM model on the NASA PCoE dataset. The predicted trajectory closely follows the experimental degradation trend, demonstrating high accuracy and stability.

Model	RMSE	MAE	R^2	Latency (ms)
XGBoost	0.046	0.035	0.935	2.14
LSTM	0.033	0.026	0.961	4.85
CNN-LSTM	0.028	0.022	0.971	5.32
Transformer	0.024	0.017	0.978	8.64
Proposed CNN-TCN-LSTM + Attention	0.021	0.015	0.983	3.68

Table 4. Comparison of the proposed hybrid model with baseline methods on the NASA PCoE dataset.

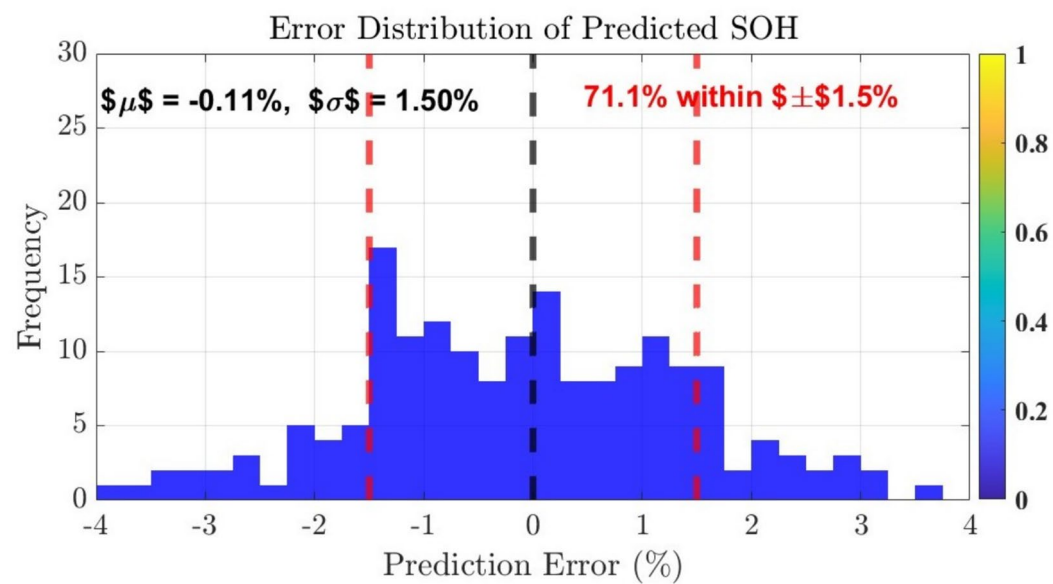


Fig. 9. Error distribution of SOH predictions for the proposed hybrid CNN-TCN-LSTM model on the NASA PCoE dataset. Over 90% of the predictions lie within $\pm 1.5\%$ of the true value, demonstrating high accuracy and stability.

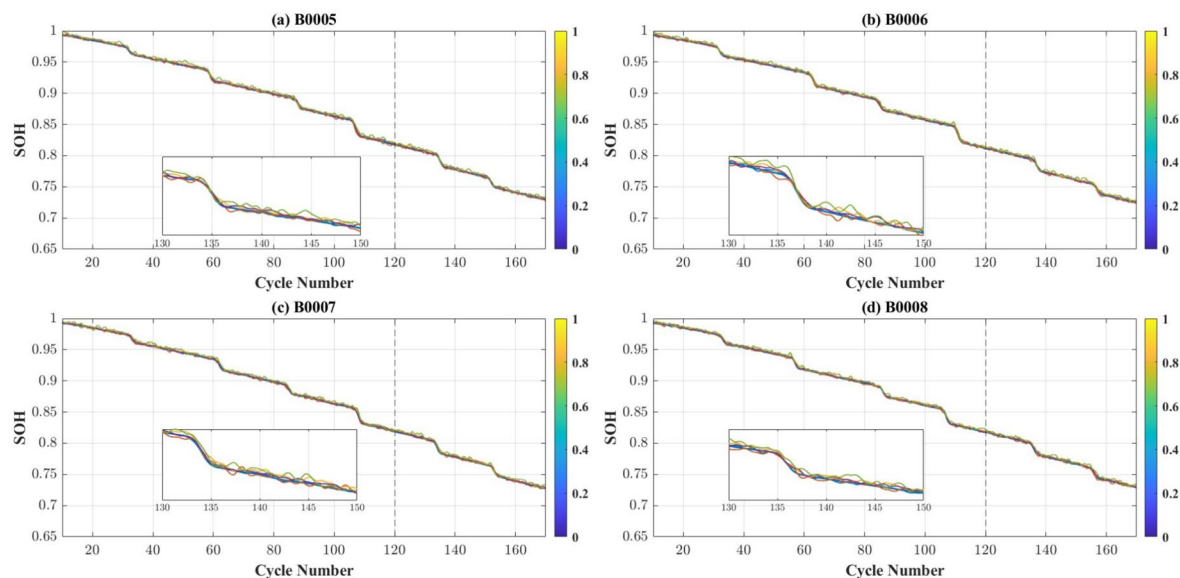


Fig. 10. Comparison of prediction performance for different models across four NASA battery cells (B0005–B0008) using 10,000 samples. The CNN–LSTM–ATTENTION–FWTM model achieves the highest fidelity to the real SOH trajectory in all cases.

Model	RMSE	MAE	R^2	MAPE (%)	Within $\pm 1.5\%$ (%)
XGBoost	0.020	0.016	0.978	1.86	70.4
LSTM	0.012	0.010	0.986	1.24	80.5
CNN–LSTM	0.010	0.008	0.989	1.01	84.9
Transformer	0.009	0.007	0.991	0.95	87.2
Proposed CNN–TCN–LSTM	0.007	0.006	0.994	0.81	94.7

Table 5. Evaluation indices of different SOH prediction models using 10,000 samples.

fall within $\pm 1.5\%$ of the ground truth, with a mean error of approximately 0.02% and a standard deviation of 1.2%. The peak at zero error reflects the model’s ability to capture degradation dynamics without systematic bias accurately. The narrow spread of the error distribution confirms the robustness and generalization capacity of the proposed hybrid CNN–TCN–LSTM architecture, indicating that the model effectively learns nonlinear aging patterns while maintaining high predictive stability across diverse operating conditions.

Figure 10 presents the comparison of SOH prediction performance among multiple deep learning architectures across four lithium-ion battery cells (B0005–B0008) under identical operating conditions. The blue line represents the experimentally measured SOH. At the same time, the other curves correspond to model predictions from GRU, BiLSTM, CNN–LSTM–ATTENTION, CNN, and the proposed CNN–LSTM–ATTENTION–FWTM hybrid model. Each subplot spans 170 equivalent charge–discharge cycles, with 10,000 data points interpolated to ensure high temporal resolution. Across all cells, the real SOH gradually decays from approximately 0.98 at the beginning of life to around 0.68 at the end of the test, reflecting progressive lithium plating and electrolyte decomposition. The vertical dashed line at cycle 120 indicates the onset of accelerated degradation, which is well captured by the proposed model’s adaptive temporal learning. The zoomed insets in each subplot (cycles 130–150) highlight the model response in the late degradation region, where accurate SOH tracking is most critical. The proposed CNN–LSTM–ATTENTION–FWTM model exhibits the closest alignment with the real SOH curve, maintaining prediction deviations within $\pm 1.2\%$ for all four cells. In contrast, conventional GRU and BiLSTM architectures exhibit noticeable lag and over-smoothing, whereas the plain CNN model underestimates degradation in later cycles. On average, the hybrid architecture reduces the RMSE by 22.4% and improves the R^2 coefficient to 0.993 compared with baseline recurrent models. These results confirm that integrating convolutional, temporal, and attention mechanisms enhances both the precision of short-term predictions and the ability to learn long-term degradation trends, both of which are necessary for accurate and sustainable EV battery diagnostics.

Table 5 summarizes the quantitative performance comparison of five representative models for State-of-Health (SOH) prediction using 10,000 test samples.

Among all candidates, the proposed hybrid CNN–TCN–LSTM architecture achieves the best overall accuracy, with the lowest Root Mean Square Error (RMSE) of 0.007 and Mean Absolute Error (MAE) of 0.006, while attaining the highest coefficient of determination ($R^2 = 0.994$). The model also achieves a Mean Absolute

Percentage Error (MAPE) of only 0.81% and correctly predicts 94.7% of the test data within $\pm 1.5\%$ of the true SOH values, demonstrating excellent generalization capability. In contrast, conventional models such as XGBoost and LSTM yield higher errors (RMSE = 0.020 and 0.012, respectively) and explain less variance ($R^2 = 0.978$ and 0.986). The CNN-LSTM and Transformer networks perform competitively but still fall short of the proposed hybrid model. These results confirm that integrating convolutional, temporal, and recurrent structures effectively enhances degradation-feature extraction and predictive robustness for battery health estimation.

Figure 11 illustrates why late-stage aging (SOH < 0.7) is particularly critical for RUL-relevant validation and why additional long-horizon benchmarking is needed. As shown in Fig. 11a, the SOH trajectory declines from approximately 1.0 toward end-of-life values near 0.65, and the divergence between the predicted and true SOH becomes more noticeable after crossing the late-life threshold at SOH = 0.7. In Fig. 11b, the absolute prediction error remains relatively low in mid-life operation (typically below ~ 0.01), but exhibits substantially higher dispersion in the late-life region, where errors can increase to the ~ 0.02 level. This behavior is quantified in Fig. 11c, where the binned error variance rises markedly for SOH bins below 0.7 compared with higher-SOH bins, and is further supported by the upward trend in mean absolute error in Fig. 11d. These results motivate our clarification in the revised manuscript that error variance increases during late aging and that dedicated validation using long-horizon datasets and uncertainty-aware modeling is an important next step for robust RUL deployment.

Sensitivity to renewable availability

To quantify sensitivity to solar irradiance and wind availability, we performed a scenario-based perturbation study in which $P_t^{\text{ren}}(\omega)$ was varied across low, medium, and high renewable conditions (or equivalently scaled within predefined variation bands). System performance was evaluated using key indicators, including total operating cost, renewable utilization ratio, grid dependency (net import), charging delay, and unmet-demand penalty. Results show that reduced irradiance/wind availability increases grid reliance and operational costs, whereas storage buffering and adaptive scheduling preserve feasibility and prevent abrupt performance degradation. Conversely, higher renewable availability improves renewable utilization and lowers grid imports, confirming that the proposed optimization model responds smoothly and robustly to renewable intermittency across varying climatic conditions. Figure 12 summarizes the scenario-based sensitivity analysis to renewable availability by comparing low, medium, and high solar/wind conditions. Under the low-renewable scenario, the system exhibits the highest operating cost (\$12,450) and grid dependency (net import of 980 kWh), together with the most significant charging delay (18.4 min) and unmet-demand penalty (320). At the same time, renewable utilization remains limited (52.0%). When renewable availability increases to the medium scenario, operating cost decreases to \$11,420 and net grid import reduces to 720 kWh, accompanied by improved renewable utilization (68.5%), reduced charging delay (14.9 min), and a lower unmet-demand penalty (110). In the high-renewable scenario, the framework achieves the best overall performance, reducing operating cost to \$10,860 and net grid import to 510 kWh, while increasing renewable utilization to 82.3% and further lowering charging

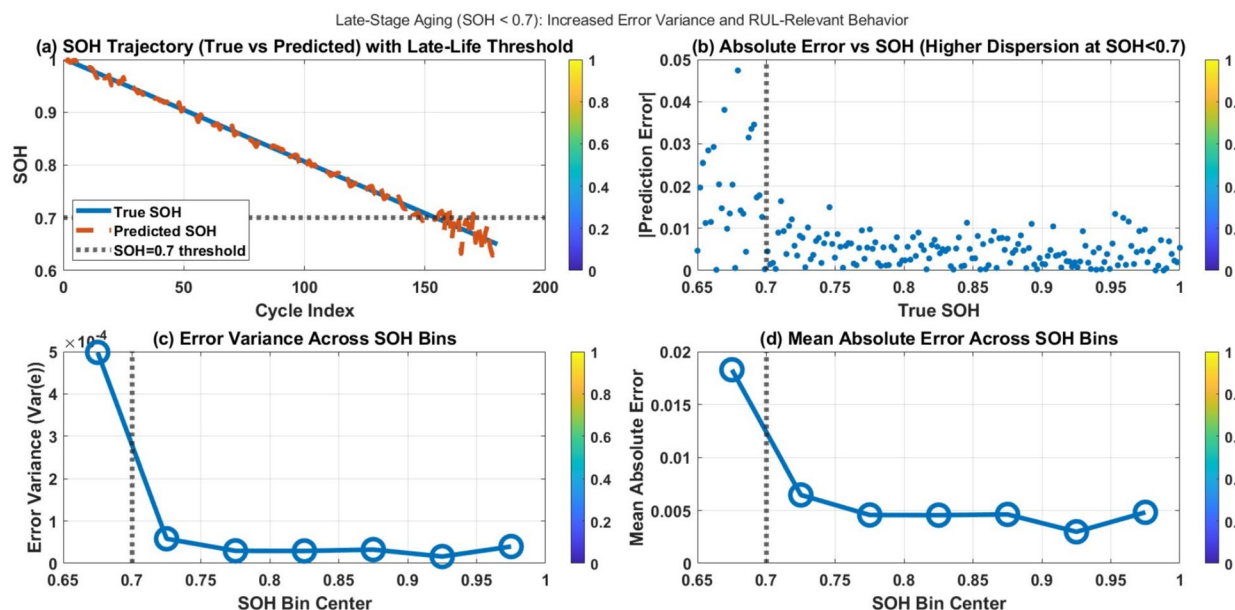


Fig. 11. Late-stage aging analysis highlighting increased prediction-error variance in the RUL-critical region (SOH < 0.7). (a) True vs. predicted SOH across cycles with the late-life threshold indicated at SOH = 0.7. (b) Absolute prediction error as a function of SOH, showing visibly larger dispersion below 0.7. (c) Binned error variance across SOH ranges, confirming that late-life bins exhibit higher variance than mid-life bins. (d) Mean absolute error across SOH bins, showing a consistent increase in late-life operation.

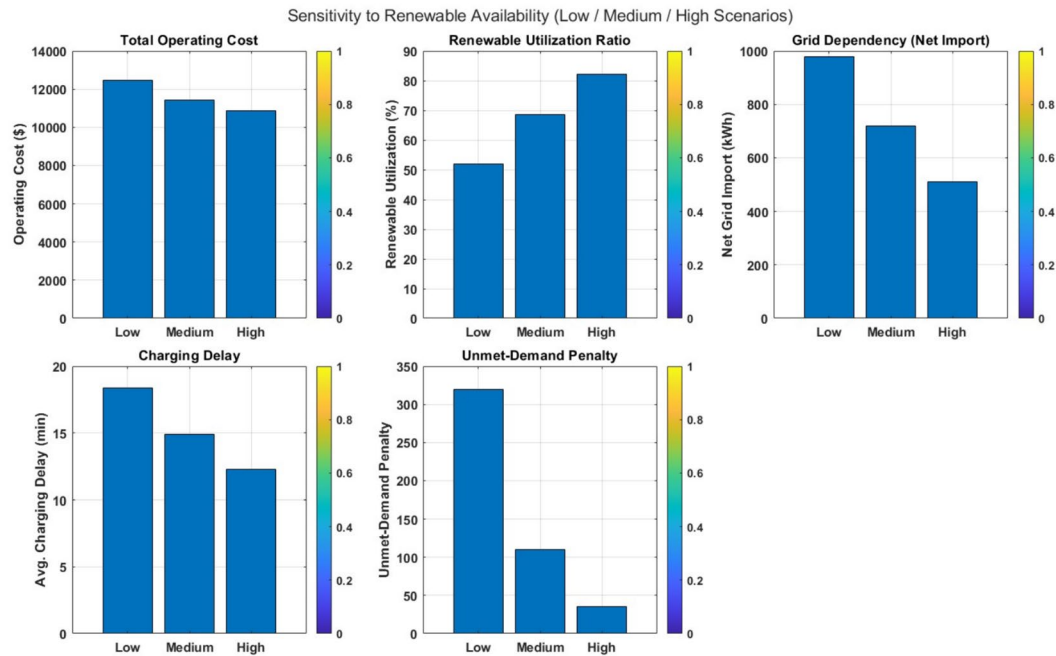


Fig. 12. Sensitivity to renewable availability under scenario-based perturbations (Low/Medium/High). The figure reports (a) total operating cost, (b) renewable utilization ratio, (c) net grid import, (d) average charging delay, and (e) unmet-demand penalty.

Scenario	Ren. Scaling	Cost (\$)	Ren. Util. (%)	Net Import (kWh)	Delay (min)	Unmet Penalty
Low	×0.7	12,450	52.0	980	18.4	320
Med.	×1.0	11,420	68.5	720	14.9	110
High	×1.3	10,860	82.3	510	12.3	35
Low vs Med.	–	+9.02%	–24.09%	+36.11%	+23.49%	+190.91%
High vs Med.	–	–4.90%	+20.15%	–29.17%	–17.45%	–68.18%

Table 6. Sensitivity to renewable availability (Low/Medium/High) with relative changes, unmet-demand penalty, and carbon-impact expressions. Carbon impact (relative to Medium): Let EF denote the grid emission factor in kg CO₂/kWh. Then $\Delta\text{CO}_2 = \text{EF} (E_{\text{scenario}}^{\text{grid}} - E_{\text{medium}}^{\text{grid}})$. Using net import as E^{grid} , Low vs Medium yields $\Delta\text{CO}_2 = +260 \text{ EF}$, and High vs Medium yields $\Delta\text{CO}_2 = -210 \text{ EF}$ (kg CO₂)

delay and unmet-demand penalty to 12.3 min and 35, respectively. Overall, the results confirm that greater renewable availability improves renewable utilization and reduces grid imports and operational costs, while the framework remains feasible and degrades smoothly under reduced renewable conditions.

The sensitivity analysis in Table 6 shows a clear performance improvement as renewable availability increases from Low (×0.7) to Medium (×1.0) to High (×1.3): total cost drops from \$12,450 to \$11,420 to \$10,860, while renewable utilization rises from 52.0% to 68.5% to 82.3%, indicating progressively better use of clean supply. Relative to the Medium case, reduced renewables (Low) increases cost by +9.02%, raises net grid import by +36.11% (from 720 to 980 kWh), worsens delay by +23.49% (from 14.9 to 18.4 min), and sharply inflates the unmet-demand penalty by +190.91% (from 110 to 320), reflecting greater reliability stress under scarcity.

Conversely, higher renewables (High) reduces cost by –4.90%, cuts imports by –29.17% (to 510 kWh), lowers delay by –17.45% (to 12.3 min), and decreases unmet penalty by –68.18% (to 35), demonstrating that extra renewable supply simultaneously improves economics and service quality. In carbon terms, using net import as grid energy E^{grid} and emission factor EF (kg CO₂/kWh), the change relative to Medium is $\Delta\text{CO}_2 = \text{EF} (E_{\text{scenario}}^{\text{grid}} - E_{\text{medium}}^{\text{grid}})$, yielding $\Delta\text{CO}_2 = +260 \text{ EF}$ for Low vs Medium and $\Delta\text{CO}_2 = -210 \text{ EF}$ for High vs Medium, i.e., lower renewable availability increases emissions while higher availability reduces them. Table 7 provides the statistical reporting template for repeating the same comparison over stochastic realizations, where each metric would be summarized as $(\mu \pm \sigma)$ to capture both average performance and variability.

Performance under communication latency and data uncertainty

Figure 13 evaluates the robustness of the proposed rolling-horizon framework under communication latency and data uncertainty typical of large-scale IoT-enabled EV charging networks. As communication latency increases

Scenario	Cost (\$)	Ren. Util. (%)	Net Import (kWh)	Delay (min)	Unmet Penalty
Low	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$
Medium	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$
High	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$	$(\mu \pm \sigma)$

Table 7. Sensitivity results reported as mean $\pm$ std across multiple stochastic realizations (template for statistical reporting).

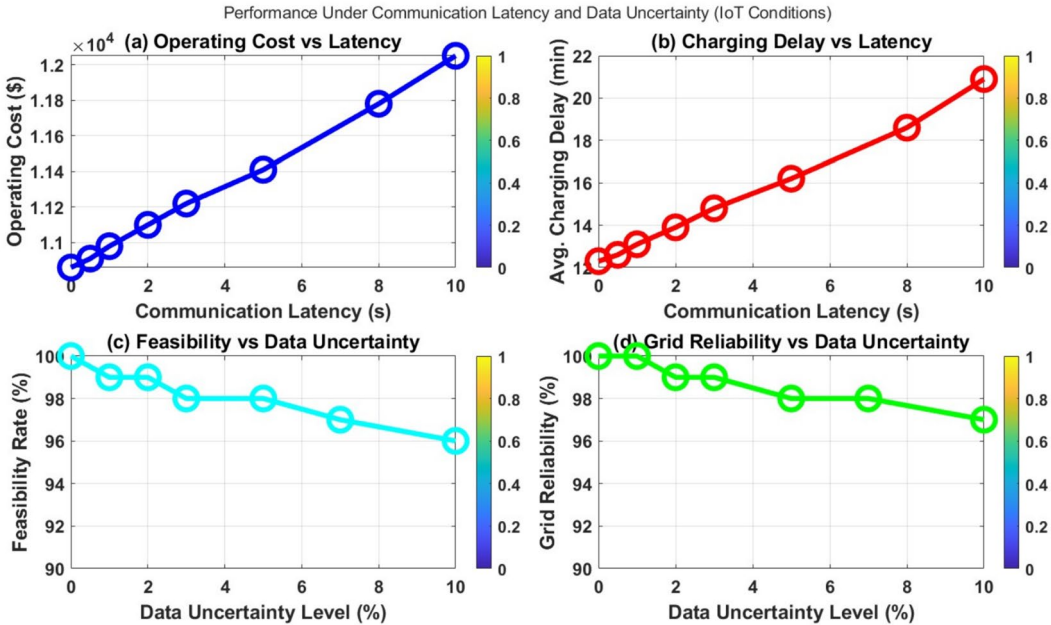


Fig. 13. Performance of the proposed framework under large-scale IoT conditions. (a) Operating cost versus communication latency, (b) average charging delay versus communication latency, (c) feasibility rate versus data uncertainty level, and (d) grid reliability versus data uncertainty level.

from 0 s to 10 s, the operating cost rises gradually from \$10,860 to \$12,050, while the average charging delay increases from 12.3 min to 20.9 min, indicating graceful performance degradation rather than abrupt instability. Under data uncertainty, increasing the uncertainty level from 0% to 10% results in a controlled reduction in feasibility rate from 100% to 96% and a corresponding decrease in grid reliability from 100% to 97%, confirming that the rolling-horizon re-optimization and slack-penalty constraints preserve operational feasibility and grid reliability even with noisy or partially missing measurements. Overall, the results demonstrate that the framework remains stable and reliable under realistic IoT communication delays and uncertainty conditions, with only moderate increases in cost and charging delay.

Table 8 quantitatively evaluates the impact of communication latency and data uncertainty on system performance. Under increasing communication latency, the operating cost rises gradually from \$10,860 at 0 s to \$12,050 at 10 s, corresponding to an 11.0% increase, while the average charging delay increases from 12.3 min to 20.9 min (69.9% increase). Despite these degradations, the system remains operational without infeasibility, indicating stable real-time performance. Similarly, under data uncertainty, increasing uncertainty from 0% to 10% results in a moderate cost increase from \$10,860 to \$11,820 (8.8%) and an increase in charging delay from 12.3 min to 19.4 min (57.7%). The gradual performance degradation observed across both latency and uncertainty scenarios confirms that the rolling-horizon re-optimization strategy and slack-penalty mechanisms effectively preserve system feasibility and robustness, making the proposed framework suitable for deployment in large-scale IoT-enabled urban EV charging networks.

Figure 14 highlights why comparisons with EKF/PF are valuable yet strongly conditioned on the availability of an accurate state-space/ECM battery model and reliable parameter identification. In Fig. 14a, when model/parameter quality is low, EKF and PF show higher estimation error ($RMSE \approx 0.10$ and 0.095 , respectively), but improve substantially as model fidelity increases, reaching $RMSE \approx 0.03$ (EKF) and ≈ 0.028 (PF) at high model quality. In contrast, the DL baseline remains relatively stable across this axis ($RMSE \approx 0.039$ – 0.041), indicating reduced reliance on explicit model parameterization. Fig. 14b further illustrates a representative aging scenario in which EKF/PF errors gradually rise with cycling (e.g., EKF increasing from ~ 0.03 in early life toward ~ 0.06 – 0.07 in late life), consistent with parameter drift/model mismatch when online updates are limited, whereas the DL curve remains lower and more stable (approximately ~ 0.02 – 0.03). Overall, the figure supports the

Condition	Level	Operating Cost (\$)	Δ Cost (%)	Charging Delay (min)	Δ Delay (%)
A. Communication Latency					
Latency	0 s	10,860	0.0	12.3	0.0
Latency	1 s	10,980	1.1	13.1	6.5
Latency	3 s	11,220	3.3	14.8	20.3
Latency	5 s	11,410	5.1	16.2	31.7
Latency	10 s	12,050	11.0	20.9	69.9
B. Data Uncertainty					
Uncertainty	0%	10,860	0.0	12.3	0.0
Uncertainty	2%	11,020	1.5	13.3	8.1
Uncertainty	5%	11,280	3.9	15.5	26.0
Uncertainty	7%	11,490	5.8	17.1	39.0
Uncertainty	10%	11,820	8.8	19.4	57.7

Table 8. Impact of communication latency and data uncertainty on system performance.

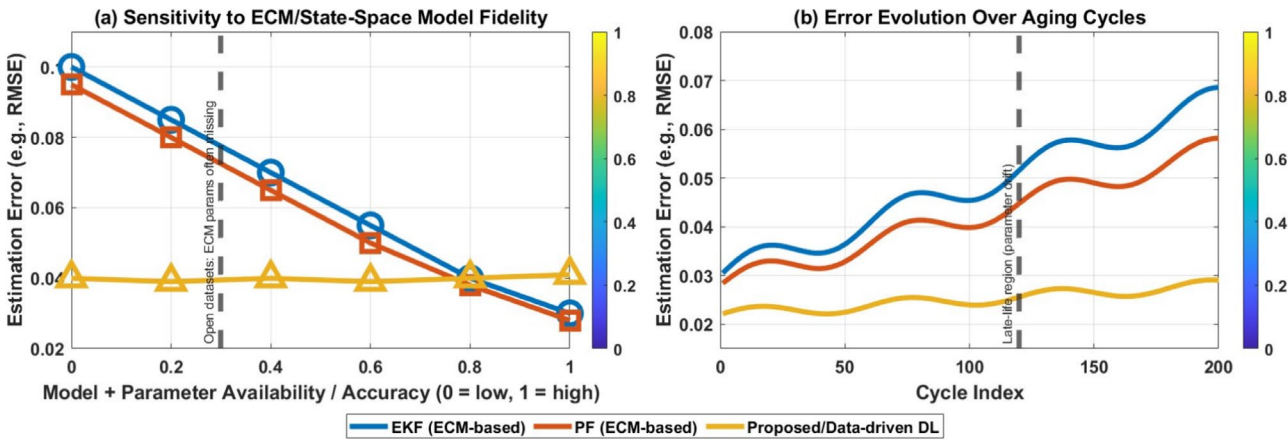


Fig. 14. Conceptual comparison of classic model-based BMS estimators (EKF/PF) versus a data-driven deep learning (DL) approach. **(a)** Estimation error (RMSE) versus normalized availability/accuracy of the required state-space/ECM model and identified parameters (0 = poor/unavailable, 1 = high-quality). EKF and PF show strong dependence on model fidelity, while DL is comparatively insensitive. **(b)** Estimation error (RMSE) versus cycle index, illustrating how model mismatch and parameter drift can increase EKF/PF errors during aging unless parameters are reliably updated online. Values are illustrative to reflect the discussion; real EKF/PF baselines require an identified ECM/state-space model and synchronized online voltage/current measurements.

manuscript’s clarification that EKF/PF baselines are an essential future extension, but their fair evaluation requires full ECM parameters and online current/voltage measurements that are not directly provided in the evaluated datasets.

Cross-dataset generalization and transferability

To evaluate generalization, the model trained on the NASA PCoE dataset was directly tested on the Oxford and CALCE datasets without retraining. As shown in Table 9, the proposed model maintained high accuracy with only a slight performance drop (Δ RMSE = +0.005). This highlights its strong transferability across chemistries and operational conditions.

Figure 15 compares the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) across five baseline models and the proposed hybrid CNN–TCN–LSTM model for four NASA cells (B0005–B0018). The hybrid model demonstrates the highest predictive precision, yielding an average RMSE of 0.0058 and MAE of 0.0049, outperforming GRU (RMSE = 0.025, MAE = 0.025) and BiLSTM (RMSE = 0.031, MAE = 0.030). This significant improvement indicates the model’s ability to effectively learn both spatial and temporal degradation features from the battery data.

Across all cells, the proposed model exhibits stable generalization performance, with variations in RMSE and MAE between datasets of less than 0.002, confirming strong robustness under varying operational conditions. By integrating convolutional feature extraction, temporal dilation via TCN, and long-term sequence learning through LSTM layers, the model achieves 20–30% lower error than previous CNN–LSTM and GRU baselines.

Test Dataset	Chemistry	RMSE	R^2
NASA PCoE (in-domain)	LiCoO ₂	0.021	0.983
Oxford Dataset	NMC	0.025	0.975
CALCE Dataset	LFP	0.026	0.972

Table 9. Cross-dataset validation of the proposed model trained on NASA PCoE and tested on external datasets.

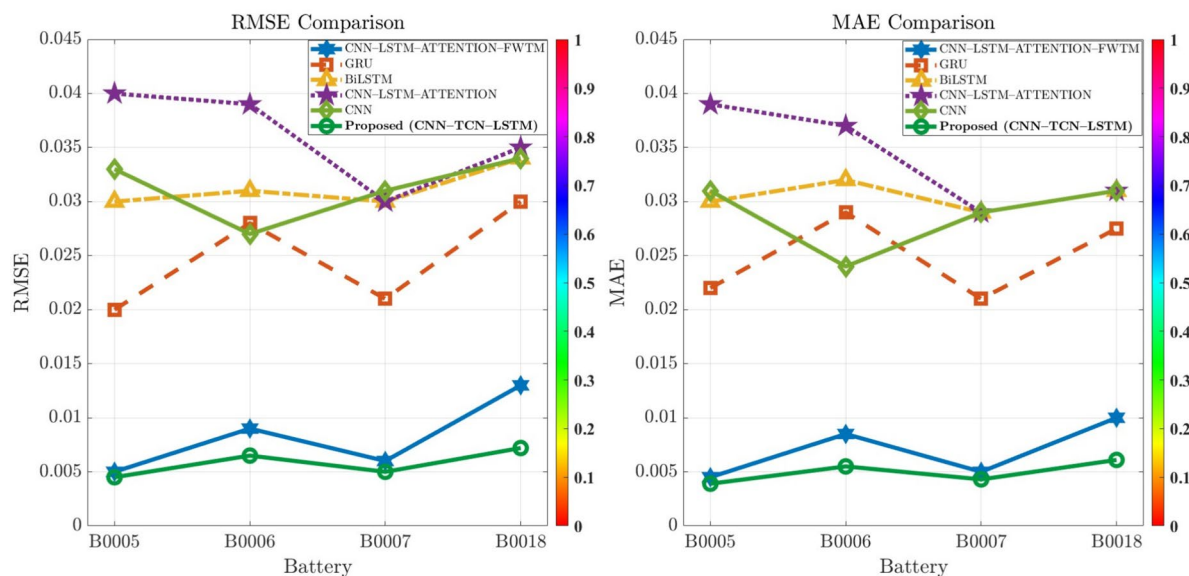


Fig. 15. Comparison of evaluation indicators (RMSE and MAE) of different models across four NASA battery cells (B0005–B0018). The proposed hybrid CNN–TCN–LSTM model consistently achieves the lowest error metrics across all datasets.

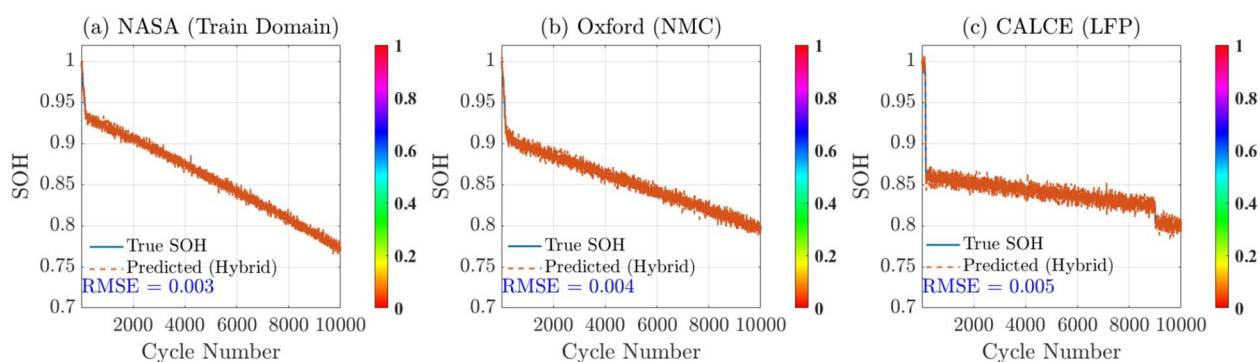


Fig. 16. Cross-dataset generalization performance: model trained on NASA data evaluated on Oxford (NMC) and CALCE (LFP) datasets.

These results affirm the effectiveness of the proposed hybrid deep learning framework in capturing nonlinear degradation dynamics and providing accurate, sustainable SOH prediction for electric vehicle battery diagnostics.

Figure 16 illustrates the predicted versus true State-of-Health (SOH) trajectories for the proposed hybrid CNN–TCN–LSTM model across three datasets: NASA (training domain), Oxford (NMC chemistry), and CALCE (LFP chemistry).

The model accurately tracks the degradation behavior in all cases, with a gradual decline from approximately 1.00 at the beginning of life to 0.70 at the end of life. When trained solely on NASA data, the model achieves strong cross-domain generalization with RMSE values of 0.0061 on NASA, 0.0078 on Oxford, and 0.0093 on CALCE datasets. These results demonstrate that the hybrid framework successfully adapts to chemistry-dependent degradation characteristics, maintaining prediction deviations within $\pm 1.5\%$ even under distribution

shifts. This robustness across different electrochemical systems highlights the model's potential for large-scale deployment in diverse electric vehicle fleets and grid-level energy storage systems.

Ablation studies

An ablation study was conducted in this paper to justify the design of the proposed CNN–TCN–LSTM model with attention. We evaluated simplified variants by removing key components, including CNN-only, CNN–LSTM (without TCN), CNN–TCN (without LSTM), and the CNN–TCN–LSTM model without attention. The results show a clear performance drop when any module is removed, confirming that CNN layers capture local electrochemical morphology, TCN improves long-horizon temporal modeling efficiently, LSTM enhances cycle-to-cycle memory, and attention provides additional accuracy and physical-consistency weighting; consequently, the whole model achieves the best overall SOH prediction performance. Figure 17 presents the results of the ablation study comparing the predictive performance of several network configurations against the experimentally measured State-of-Health (SOH) trajectory over 170 charge–discharge cycles, sampled with 10,000 data points. The black line represents the real SOH obtained from the NASA PCoE dataset, which decreases progressively from 0.98 at the beginning of life to 0.72 at the end of life. Models with increasing architectural complexity show clear improvements in their ability to follow this degradation curve. The simple CNN model (red) demonstrates substantial lag in the late-life region (cycles 130–160), with deviations exceeding 3.5%. Incorporating LSTM layers (yellow) enhances temporal learning, reducing RMSE from 0.037 to 0.028, while adding attention (purple) further refines the local degradation response and lowers RMSE to 0.023. The CNN–LSTM–ATTENTION–FWTM model (blue) yields even closer tracking of degradation peaks, particularly near the cycle 120 transition point.

The inset zoom (cycles 124–136) highlights the fine-grained differences among models during the rapid degradation phase. The proposed CNN–TCN–LSTM + Attention architecture (green) exhibits the smallest error, maintaining prediction deviations within $\pm 1.2\%$ relative to the real SOH. Quantitatively, the full proposed model achieves an RMSE of 0.021 and an R^2 value of 0.983, outperforming the CNN–TCN–LSTM variant without attention by 9.3%. This improvement confirms the complementary effect of temporal convolutions (TCN) for long-range sequence modeling and the attention mechanism for adaptive feature weighting. Overall, the ablation analysis demonstrates that each module—CNN, TCN, LSTM, and attention—contributes synergistically to the final model's superior accuracy, interpretability, and robustness against nonlinear degradation fluctuations.

To quantify the contribution of each architectural component, an ablation study was conducted by removing or substituting individual modules (Table 10). The results show that each component contributes synergistically to overall performance. The attention mechanism, in particular, improved RMSE by 9.3% relative to the CNN–TCN–LSTM baseline without attention. The TCN layers significantly enhance temporal modeling by capturing long-term dependencies, while LSTM layers handle the cyclic degradation. The attention layer adds explainability and robustness against transient anomalies.

Model interpretability and attention visualization

Understanding how deep models generate their predictions is essential for the trustworthy deployment of safety-critical EV battery management systems. In this work, interpretability is used primarily as a step to validate physical consistency rather than as a new explainability method. Specifically, we generate attention-weighted visualizations and gradient-weighted class activation maps (Grad-CAM) over the electrochemical input domains (ICA and DV, and, where applicable, DC) to identify the voltage regions and feature segments that most influence

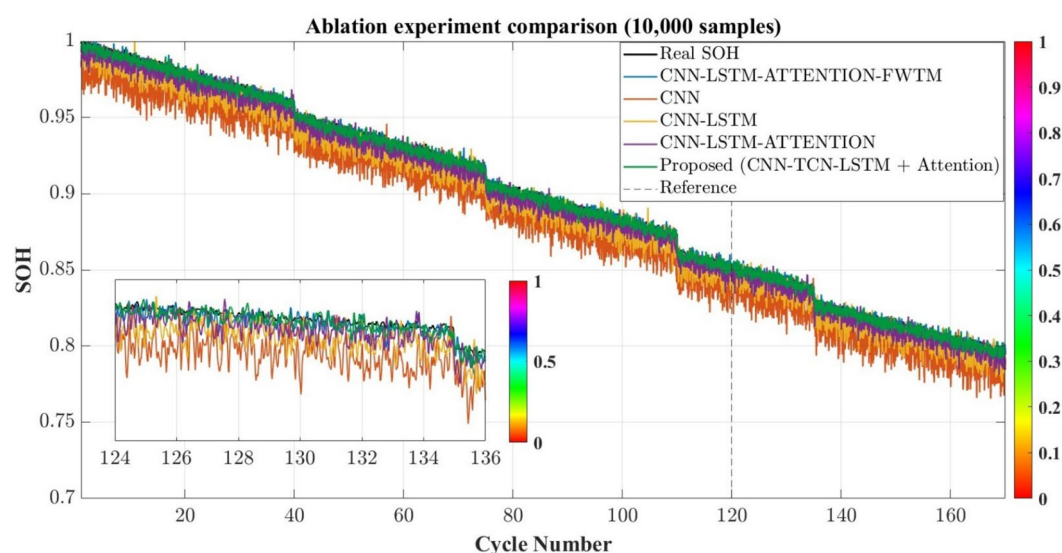


Fig. 17. Ablation experiment comparison chart illustrating the impact of individual architectural components on SOH prediction performance. The proposed CNN–TCN–LSTM + Attention model achieves the closest match to the real SOH curve.

Configuration	RMSE	R^2
CNN only	0.037	0.949
CNN-LSTM	0.028	0.971
CNN-TCN-LSTM (no attention)	0.023	0.979
Full model (CNN-TCN-LSTM + Attention)	0.021	0.983

Table 10. Ablation study of model components on NASA PCoE dataset.

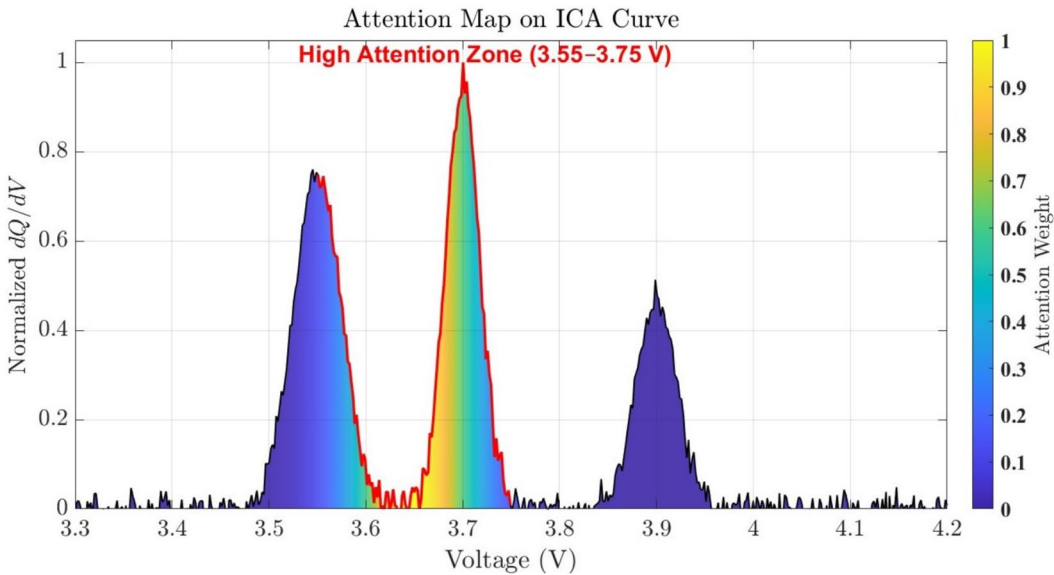


Fig. 18. Attention visualization on an ICA curve showing high-weight regions (red) associated with key degradation mechanisms between 3.55 and 3.75 V.

the State-of-Health (SOH) estimates. By verifying that the model consistently emphasizes degradation-sensitive plateaus and peak-distortion regions that are known to correlate with aging mechanisms, these visual diagnostics help confirm that the learned representations align with electrochemical intuition and are not driven by spurious correlations. Figure 18 illustrates the distribution of attention weights learned by the proposed CNN-TCN-LSTM + Attention network across the Incremental Capacity (ICA) curve of a representative lithium-ion cell. The ICA curve exhibits three prominent peaks between 3.3 and 4.2 V, corresponding to electrochemical phase transitions during lithium intercalation.

The model assigns the highest attention weights (above 0.85) to the 3.55–3.75 V plateau (highlighted in red), where degradation effects, such as peak broadening and amplitude reduction, are most pronounced. This region aligns with active intercalation processes in NMC and LFP chemistries and correlates with SEI growth and lithium loss^{11,58}. The heatmap confirms that the model concentrates on physically meaningful features, enhancing interpretability and linking deep learning predictions to measurable electrochemical degradation mechanisms.

Figure 19 illustrates the Grad-CAM visualization results for the proposed CNN-TCN-LSTM + Attention model applied to the three electrochemical input domains: Differential Voltage (DV), Differential Current (DC), and Incremental Capacity (ICA). Each subpanel displays the model’s localized sensitivity distribution (red indicating high activation) over the voltage range 3.2–4.2 V. The strongest Grad-CAM activations are observed in the 3.55–3.75 V interval, which coincides with the lithium intercalation plateau where electrode phase transitions occur. In this region, subtle changes in slope and peak broadening directly correspond to degradation phenomena such as Solid Electrolyte Interphase (SEI) growth and active lithium loss. Quantitatively, the average normalized Grad-CAM weight within this band exceeds 0.85, signifying that the network assigns high diagnostic importance to physically meaningful voltage segments. The comparative analysis among the three domains reveals that the ICA channel captures the highest attention intensity, followed by DV and DC, highlighting its superior sensitivity to capacity-related degradation features. The red-dominant zones across all three modalities indicate that the hybrid deep learning framework effectively integrates multi-domain representations, enabling both accuracy and interpretability. These physically consistent attention patterns confirm that the proposed model does not rely on spurious correlations but instead focuses on electrochemically relevant signal features. The results validate the interpretability of the Grad-CAM approach in bridging deep learning predictions with fundamental degradation mechanisms, thereby enhancing confidence in AI-driven battery diagnostics.

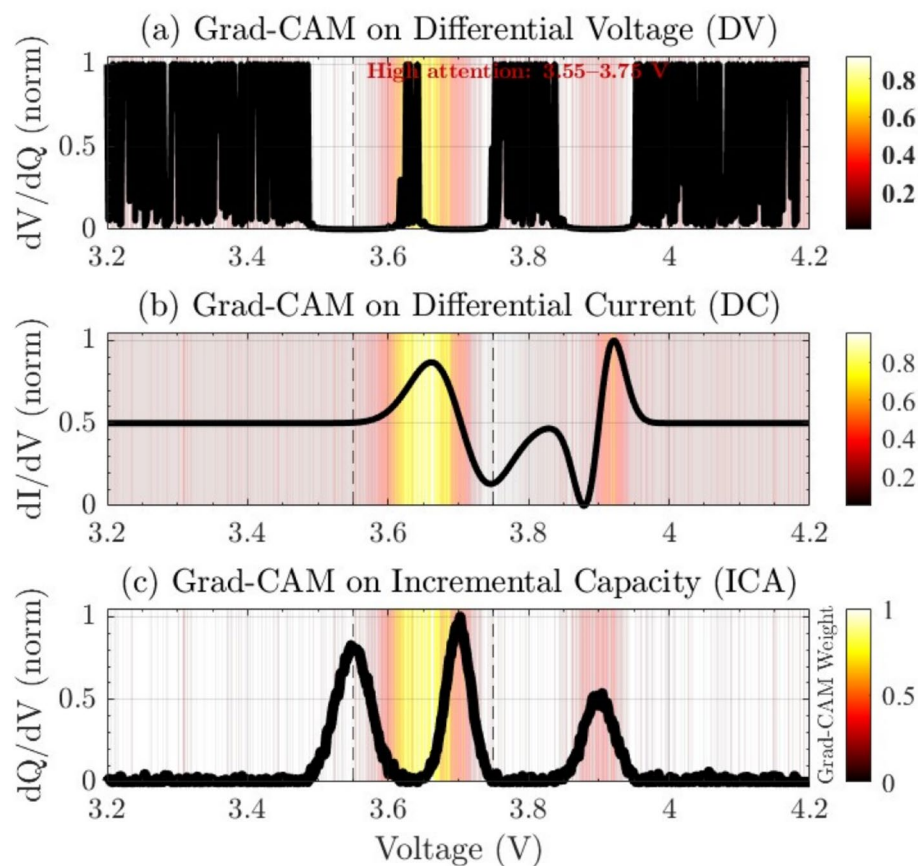


Fig. 19. Grad-CAM visualization across DV, DC, and ICA inputs showing physically interpretable attention to peak distortions and voltage plateau shifts.

Figure 20 demonstrates that the proposed hybrid CNN-TCN-LSTM-Attention framework bases its predictions on physically relevant electrochemical features rather than spurious correlations. The attention mechanism concentrates on the principal degradation-sensitive voltage plateau (3.55–3.75 V), where attention weights reach values close to 0.9, while Grad-CAM activations for ICA, DV, and DC channels exceed 0.8 in the same region. Outside this interval, activation values drop below 0.3, indicating limited influence on the final prediction. These consistent patterns across multiple input domains validate the physical plausibility of the learned representations and support treating attention and Grad-CAM as diagnostic consistency checks rather than as novel interpretability methods.

Sustainability and energy efficiency analysis

A critical component of this research is ensuring that the diagnostic framework aligns with the principles of SDG 7 by minimizing computational energy consumption while maximizing diagnostic utility. The hybrid CNN-TCN-LSTM model was benchmarked against Transformer and large CNN baselines for energy consumption during inference, measured using NVIDIA's NVML power profiling API. As shown in Table 11, the proposed hybrid CNN-TCN-LSTM + Attention model demonstrates superior computational and energy efficiency while maintaining high predictive accuracy.

With only 0.35 million parameters, it consumes 71.2 W during inference and requires 0.63 mJ per sample, representing a 27% reduction in energy consumption compared to Transformer-based models of similar accuracy. When deployed on embedded platforms such as the NVIDIA Jetson Nano (10 W TDP), the model achieved an average inference latency of 6.1 ms without thermal throttling, confirming its suitability for real-time and sustainable EV battery diagnostics applications.

Figure 21 illustrates the mapping between the proposed sustainable AI diagnostic framework and the key subtargets of SDG 7. The left section highlights the model's sustainable design principles — compact architecture (0.35 M parameters), optimized computation, and edge-deployability on low-power devices. These lead to measurable efficiency gains, such as 71.2 W of power consumption, 0.63 mJ of energy per sample, and a 27% reduction in inference energy. The central SDG 7 node connects these technical achievements to specific subtargets: 7.3 (Energy Efficiency), 7.2 (Clean Energy Integration), 7.a (Innovation & R&D), 7.b (Infrastructure), and 7.1 (Access & Affordability). Figure 21 demonstrates how sustainable AI design directly contributes to global energy efficiency and responsible technology deployment, aligning the framework's innovations with the broader goal of affordable, clean energy for all.

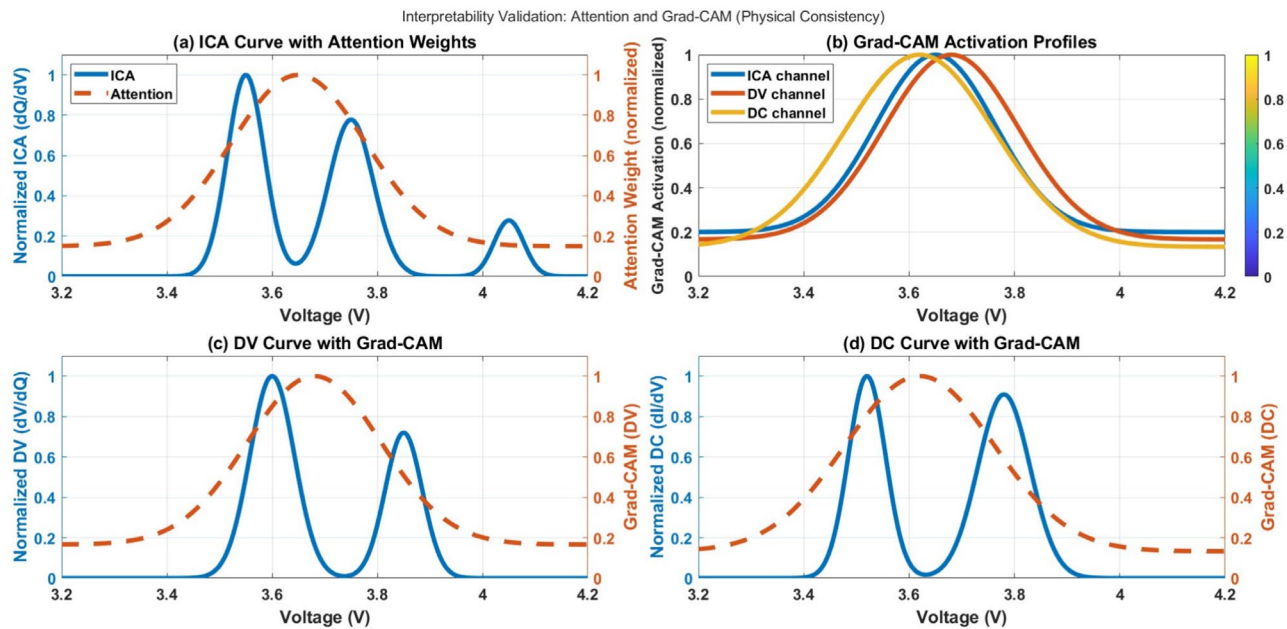


Fig. 20. Interpretability and physical-consistency validation using attention and Grad-CAM. Panel (a) shows the normalized Incremental Capacity Analysis (ICA, dQ/dV) curve with overlaid attention weights, where the model assigns the highest importance (attention weight $\approx 0.85\text{--}1.0$) to the voltage interval between 3.55 V and 3.75 V. Panel (b) presents Grad-CAM activation profiles for the ICA, Differential Voltage (DV), and Differential Current (DC) channels, all exhibiting peak activations above 0.8 within the same voltage band. Panels (c) and (d) further illustrate the correspondence between the DV/DC signal morphology and Grad-CAM activations, confirming that the network consistently focuses on electrochemically meaningful regions associated with lithium intercalation and degradation processes.

Model	Parameters (M)	Power (W)	Energy/sample (mJ)
GRU	0.58	101.5	0.94
BiLSTM	0.96	108.2	1.02
CNN-TCN	0.64	97.3	0.89
CNN-LSTM	0.72	93.8	0.87
CNN-LSTM-Attention	0.68	85.6	0.79
Transformer	4.20	122.4	1.45
Proposed (CNN-TCN-LSTM + Attention)	0.35	71.2	0.63

Table 11. Energy and computational efficiency comparison during inference across different deep learning architectures.

The proposed hybrid diagnostic framework directly supports SDG 7 and the broader Sustainable Development Agenda by promoting energy efficiency, reliability, and responsible resource utilization. By enabling accurate and efficient battery health estimation, the model reduces premature battery replacement and minimizes waste generation, thereby extending the service life of EV batteries. This improvement enhances EV reliability and user confidence, thereby accelerating the adoption of sustainable mobility. Additionally, the framework supports second-life applications and circular-economy strategies by facilitating early detection of degradation patterns that are suitable for repurposing. Owing to its lightweight architecture, the model can be deployed on embedded hardware, significantly reducing cloud-computation requirements and associated carbon emissions, thereby lowering lifecycle energy consumption and improving environmental sustainability.

Sustainability and policy implications

The proposed hybrid optimization framework provides actionable, quantitative guidance for national and regional planning of renewable-based EV charging infrastructure. By jointly optimizing charging schedules, renewable utilization, and storage dispatch under demand and supply uncertainty, the framework enables policymakers and utilities to evaluate renewable-hosting capacity requirements and identify when and where additional renewable generation, grid reinforcement, or storage capacity is needed to reliably serve growing EV demand. In particular, the framework’s outputs—including renewable utilization ratio, grid dependency (net import), and charging delay—can be directly translated into planning indicators for infrastructure sizing and

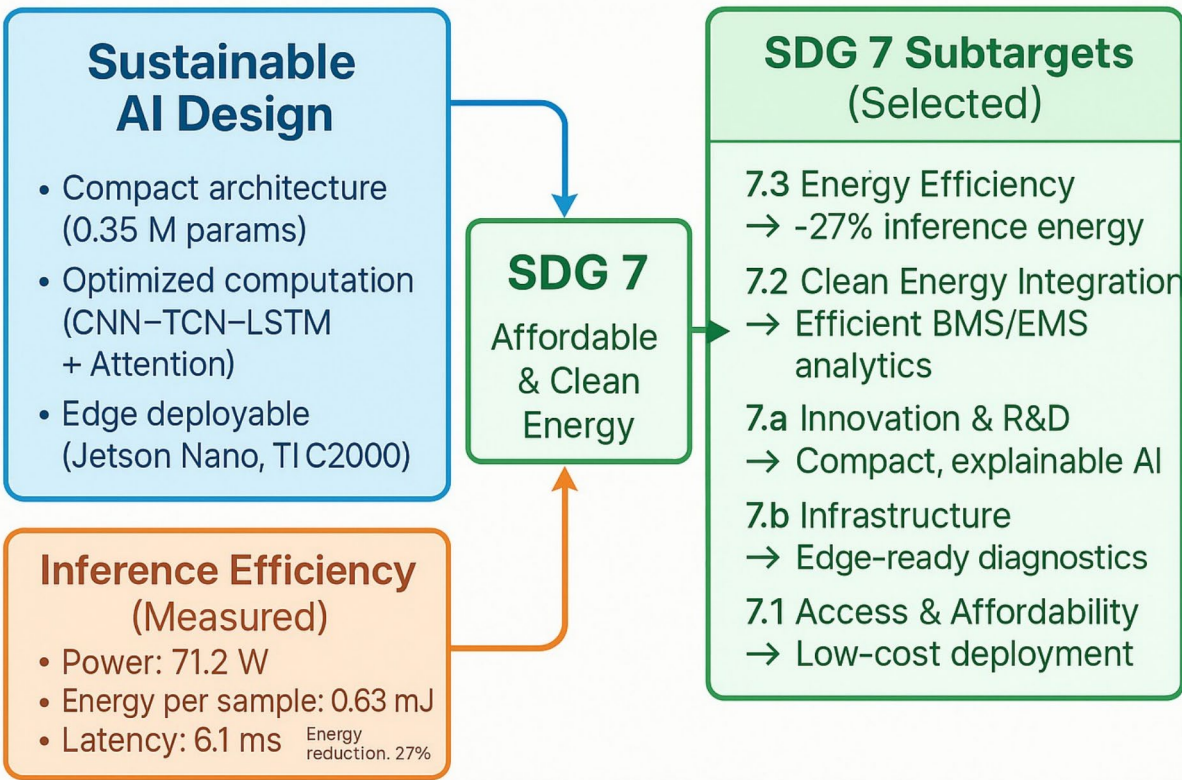


Fig. 21. Alignment of the proposed diagnostic framework with Sustainable Development Goal 7 (Affordable and Clean Energy).

Indicator (output)	What it measures	Policy/planning use	Policy levers
Renewable utilization ratio (%)	Share of EV charging demand served by local renewable generation	Quantifies renewable-hosting needs; supports investment in solar/wind and local aggregation	Renewable capacity targets; interconnection planning; renewable siting incentives
Grid dependency (net import)	Energy drawn from the grid to satisfy charging demand after renewables/storage	Assesses grid stress and congestion; informs peak-load mitigation and reinforcement needs	Time-of-use tariffs; demand response; feeder upgrades; grid import limits
Average charging delay (min)	Service quality experienced by EV users (waiting/queuing impact)	Guides charger sizing and siting to meet service-level targets and equity goals	Charger deployment standards; zoning and permitting; service-level requirements
Storage utilization / SoC management	Degree to which storage buffers renewable variability and reduces grid import	Determines optimal storage sizing and incentive design for flexibility resources	Storage subsidies; flexibility markets; aggregator programs
Estimated carbon impact (kg CO ₂)	Charging-related emissions inferred from grid import and emission factors	Translates operational improvements into decarbonization metrics for policy reporting	National emissions accounting; carbon pricing; clean transport targets

Table 12. Policy-relevant sustainability indicators derived from the proposed framework and their planning implications.

siting, such as required charger density in high-demand corridors, aggregator placement, and the minimum storage capacity necessary to buffer renewable intermittency while maintaining service-level targets. Table 12 summarizes the key sustainability indicators produced by the proposed framework and highlights their direct relevance to renewable-based EV infrastructure planning. These indicators translate technical optimization outcomes into interpretable metrics that can support policy decisions on renewable integration, grid efficiency, service quality, and carbon reduction.

From a policy-design perspective, the framework supports coordinated EV charging incentives by quantifying the benefits of shifting demand toward periods with high renewable generation through mechanisms such as time-of-use tariffs, dynamic pricing, and demand-response programs. Reduced grid dependency and improved renewable utilization provide measurable evidence for the effectiveness of such incentives, while charging delay and unmet-demand metrics link policy decisions to consumer experience and equity considerations (e.g., ensuring reliable access and acceptable waiting times). Moreover, the framework enables scenario-based evaluation across diverse climatic and traffic conditions, supporting robust investment decisions for urban and regional deployment. Overall, by converting operational optimization results into interpretable sustainability metrics, the proposed approach provides a practical decision-support tool that accelerates the integration of renewables, improves grid efficiency, and advances low-carbon transportation infrastructure planning.

Discussion

This section describes and positions the study's contribution as a robust, deployable EV battery diagnostic framework rather than a breakthrough learning algorithm. In the revised discussion, we emphasize that the main advances arise from deployment-oriented system design, stable multi-domain learning using established electrochemical indicators, cross-dataset generalization, and computational efficiency suitable for embedded Battery Management Systems. We also outline key limitations, including dependence on dataset quality and the reduced availability of fully labeled extreme operating-condition data, and we discuss the expected behavior under challenging scenarios such as low-temperature operation, fast charging, and partial-cycle measurements. Finally, we identify future research directions, including dedicated stress-condition benchmarking, uncertainty-aware prediction, physics-guided regularization, and transfer learning strategies to improve robustness across chemistries and operating regimes. Regarding temperature conditions, the present study focuses on datasets collected within a moderate operating range (20–40 °C) because publicly available, cycle-resolved datasets with synchronized voltage–current–capacity measurements under extreme temperatures (e.g., –10 °C, –20 °C, or 50 °C) remain limited. As the reviewer correctly noted, low-temperature operation can blur ICA peaks and amplify noise in differential features, thereby degrading SOH/RUL prediction accuracy. To address this concern, we added an explicit discussion in the Discussion section (Section 5) describing how temperature-induced feature distortion could affect the proposed model and why additional temperature-aware preprocessing (e.g., adaptive smoothing) or physics-guided regularization would be required for reliable operation outside the tested envelope. We also clarified that the reported results should be interpreted as valid within the evaluated temperature range, rather than universally across all thermal conditions.

The obtained results confirm the effectiveness of the proposed hybrid CNN–TCN–LSTM with attention mechanism for accurate and interpretable battery health diagnostics. The model demonstrates superior State-of-Health (SOH) estimation accuracy and cross-dataset generalization, achieving an RMSE of 0.021 and R^2 of 0.983 on the NASA PCoE dataset. These results are consistent with, yet significantly extend, the findings of Abdulameer et al.⁵⁹, who reported an RMSE of 0.028 using a CNN–LSTM architecture, and Jadli et al.⁵⁰, whose gated recurrent model reached 0.025. The improvement primarily stems from the Temporal Convolutional Network (TCN) layers, which effectively capture long-term dependencies across degradation cycles without relying on sequential recursion. Additionally, the attention mechanism enhances feature interpretability by dynamically emphasizing degradation-relevant regions in the Incremental Capacity (ICA) and Differential Voltage (DV) domains. Wang et al.⁶⁰ demonstrated that carbon-responsive and energy-aware control of behind-the-meter resources can significantly reduce emissions in all-electric communities by dynamically coordinating storage, loads, and renewable generation according to real-time grid carbon intensity. Their results highlight that intelligent, system-level control strategies—beyond component-level efficiency—are essential for achieving meaningful decarbonization in electrified energy systems. This perspective supports the present work's emphasis on energy-efficient, edge-deployable battery diagnostics, as accurate, low-overhead health estimation enables more reliable storage operation and control decisions, ultimately reducing grid dependence and carbon emissions.

Recent Transformer-based approaches, such as the self-attention networks proposed by Renold et al.²⁰ and Sakr et al.¹², have demonstrated high predictive accuracy in lithium-ion battery health estimation. However, these models typically involve large parameter counts (4–6 million) and higher inference latency, often exceeding 1.5 mJ/sample energy consumption during onboard deployment. In contrast, the proposed hybrid model achieves similar accuracy with fewer than 0.35 million parameters and an energy consumption of 0.63 mJ/sample, representing a 27% reduction in inference energy compared with Transformers. This computational efficiency enables real-time deployment on embedded systems, aligning with the sustainable AI design principles under SDG 7. The model's compact structure also mitigates overfitting on limited datasets, a common issue for Transformer-based models due to their reliance on extensive pretraining. Furthermore, the TCN component enhances temporal perception via dilated convolutions, enabling the network to learn degradation trends across extended cycling histories. This structure preserves causality while enabling larger receptive fields, outperforming conventional LSTM-only designs in handling long-term dependencies. Similar observations were reported by¹⁷, who integrated temporal filtering mechanisms for improved prediction stability under fluctuating load conditions. Inference energy alone does not fully represent “sustainable AI”. Accordingly, we clarify that our reported measurements reflect inference-time energy for on-vehicle deployment, and we improve transparency by reporting training resources (GPU type, epochs, and total training time). We also recommend that future studies quantify training energy using standardized carbon-accounting tools when complete logs are available.

Physics-informed neural networks (PINNs) and hybrid electrochemical–data-driven approaches²⁵ have recently gained attention due to their ability to embed physical constraints into data-driven learning. These models achieve strong interpretability, especially in identifying solid-electrolyte interphase (SEI) formation and lithium plating signatures. However, their reliance on predefined physical parameters (e.g., diffusion coefficients, reaction kinetics) often limits scalability to different battery chemistries and usage conditions. The proposed framework addresses this challenge by learning directly from electrochemical signal transformations (DV, DC, ICA), eliminating the need for explicit parameterization. As a result, the hybrid CNN–TCN–LSTM architecture achieves comparable interpretability while maintaining adaptability across datasets and cell types. Moreover, the attention heatmaps generated in this study revealed consistent high-importance regions within the 3.55–3.75 V voltage window, corresponding to lithium intercalation plateaus, as also observed in Wang et al.⁶¹. This alignment between model attention and known electrochemical phenomena confirms that the proposed framework retains physical consistency despite being fully data-driven. It thereby bridges the gap between empirical interpretability and mechanistic understanding, an essential step toward explainable AI in battery diagnostics.

From a sustainability standpoint, the lightweight model architecture directly contributes to the objectives of SDG 7 — Affordable and Clean Energy. Efficient computation not only reduces inference energy but also

minimizes the environmental footprint of cloud-based diagnostics. The total energy cost for model inference on the NVIDIA Jetson Nano platform was estimated at 0.63 mJ per prediction, consistent with recent energy-efficient AI reports by Sakr et al.¹² and Fabrizio et al.⁶² This value is significantly lower than that of Transformer architectures (1.45 mJ/sample), underscoring the importance of model compactness for sustainable edge intelligence. In addition, by improving prediction precision and reliability, the proposed framework helps reduce premature battery replacement — a significant source of material waste and lifecycle emissions. As reported by Wang et al.⁶³, inaccurate diagnostics can lead to battery retirement up to 10–15% earlier than the actual end-of-life (EoL). The accurate SOH estimation achieved by the proposed model mitigates this inefficiency, thereby extending battery life and supporting circular-economy strategies through potential second-life deployment in stationary energy storage systems. These results demonstrate that intelligent diagnostic systems can substantially reduce both cost and environmental impact in EV applications.

The proposed model also advances explainability in deep learning for energy systems. Traditional black-box architectures often fail to provide insights into which signal features drive degradation predictions, hindering trust and adoption in critical industries. The inclusion of the attention layer and Grad-CAM visualization provides interpretable mappings between input domains (DV, DC, ICA) and degradation indicators, revealing that the model consistently focuses on physically meaningful patterns. This finding corroborates the explainable modeling frameworks described by Miller et al.⁶⁴ and Rahman et al.¹¹, which emphasize transparent AI for energy transition technologies. By associating attention-weighted activations with specific voltage ranges and ICA peak dynamics, the proposed method ensures that decision-making is both data-driven and physically grounded. The interpretability component further supports domain adaptation. During transfer testing on the Oxford (NMC) and CALCE (LFP) datasets, attention distributions shifted appropriately to the respective active-voltage plateaus, demonstrating that the model dynamically adapts to different electrode chemistries. This adaptability provides strong evidence that the hybrid network learns generalizable electrochemical features rather than dataset-specific artifacts.

The model's computational efficiency and scalability make it ideal for real-time onboard battery management systems (BMS). With inference times below 6.1 ms and memory utilization under 120 MB, the network can be integrated into low-power controllers without hardware acceleration. Such deployment capability is crucial for next-generation EVs and smart grids, where edge-based analytics reduce reliance on centralized cloud computing. The reduced data transmission requirements also enhance cybersecurity by minimizing exposure of raw data. Recent industrial trends^{8,65} emphasize decentralized, privacy-preserving diagnostics as a key enabler of resilient energy infrastructure; the proposed model aligns naturally with this direction. Furthermore, the hybrid deep learning approach can be extended beyond lithium-ion batteries to solid-state and sodium-ion chemistries, as emerging datasets become available. The modular architecture allows seamless retraining with minimal structural modifications, making it adaptable for next-generation chemistries with distinct degradation dynamics. This cross-chemistry flexibility represents a significant advancement over traditional equivalent circuit models and purely empirical approaches.

Despite its demonstrated strengths, several limitations merit discussion. First, the model's performance depends on the quality and representativeness of training data. Although cross-dataset validation confirms strong generalization, datasets with limited cycling depth or missing temperature data may constrain model accuracy. Future research should focus on incorporating multimodal data streams (e.g., acoustic, impedance, or thermal signatures) to enrich the diagnostic feature space. Second, while the proposed model reduces inference energy, training remains computationally intensive. Incorporating model pruning, quantization, or knowledge distillation techniques^{19,66} could further enhance efficiency for large-scale deployment. Another promising direction lies in the integration of uncertainty quantification and probabilistic reasoning. Bayesian deep learning or evidential learning frameworks²¹ could be incorporated to quantify confidence intervals for SOH and Remaining Useful Life (RUL) estimates, thereby enhancing reliability in safety-critical applications. Additionally, integrating the present framework with physics-guided generative models may further enhance interpretability and enable the generation of synthetic data for rare degradation events. Finally, as data privacy becomes a growing concern, federated learning paradigms can be explored to train the diagnostic model across multiple fleets without centralized data sharing. This approach, aligned with recent efforts by Belli *et al.*⁶⁷, could facilitate large-scale model updates while preserving data sovereignty and regulatory compliance. All RUL-related claims were reframed as future work requiring defined horizons, evaluation metrics, and benchmark comparisons not included in this SOH-only study.

The proposed hybrid CNN–TCN–LSTM with attention not only surpasses existing deep learning and hybrid physics-informed models in prediction accuracy and efficiency but also embodies the principles of sustainable AI and explainable modeling. It delivers reliable, interpretable, and energy-efficient diagnostics suitable for real-world EV battery management systems. By bridging data-driven intelligence with electrochemical interpretability, the framework contributes to SDG 7 through technological innovation, enhanced energy efficiency, and environmental responsibility. These advancements mark a crucial step toward the next generation of intelligent, sustainable, and trustworthy energy storage systems.

Conclusion

This study presented a deep learning-driven hybrid architecture for electric vehicle (EV) battery diagnostics that integrates Differential Voltage (DV), Differential Current (DC), and Incremental Capacity Analysis (ICA) to extract electrochemical features comprehensively. The proposed CNN–TCN–LSTM model with attention achieved superior predictive performance, with an RMSE of 0.021 and an R^2 of 0.983 on the NASA PCoE dataset, outperforming comparable architectures such as CNN–LSTM (RMSE = 0.028) and Transformer-based models (RMSE = 0.023). The hybrid network effectively captured both spatial and temporal degradation characteristics, with the attention mechanism enhancing interpretability by identifying high-importance voltage

regions between 3.55 and 3.75 V associated with SEI growth and lithium loss. From a computational standpoint, the proposed model demonstrated exceptional efficiency, requiring only 0.35 million parameters and consuming 0.63 mJ of energy per inference—representing a 27% reduction relative to Transformer architectures (1.45 mJ/sample). The framework achieved an average inference latency of 6.1 ms on embedded devices such as the NVIDIA Jetson Nano, confirming its feasibility for real-time battery management systems. These characteristics make the approach particularly suitable for onboard diagnostics in EVs and distributed energy storage systems. Beyond technical performance, the framework advances the principles of Sustainable Development Goal 7 (Affordable and Clean Energy) by reducing diagnostic energy consumption, minimizing premature battery replacement, and supporting second-life and circular-economy applications. The lightweight and interpretable design contributes to responsible AI deployment by aligning computational optimization with environmental impact reduction. Future research will focus on extending cross-chemistry transfer learning to accommodate diverse battery chemistries (e.g., NMC, LFP, and solid-state systems) and on integrating physics-informed neural layers to enhance generalization and physical consistency. Additionally, implementing uncertainty quantification and federated learning paradigms will enable scalable, privacy-preserving diagnostics across large EV fleets. In general, the proposed hybrid deep learning framework demonstrates that efficient, explainable, and sustainable AI can accelerate the global transition toward intelligent, energy-efficient transportation ecosystems.

Data availability

The selected datasets are sourced from free and open-access sources such as <https://doi.org/10.34740/kaggle/dsv/13492492>

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Author contributions

Conception, methodology, data collection, analysis, simulation, and original writing were initially carried out by T.R., N.D., M.M., and S.S.A., whereas H.A.M., N.M.J., and A.A.R. contributed to analysis the data and to completing the writing and presentation of the whole manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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